

UNIVERSITY COLLEGE LONDON

DEPARTMENT OF COMPUTER SCIENCE

A DISSERTATION SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR
THE DEGREE OF MASTER OF SCIENCE IN EMERGING DIGITAL TECHNOLOGIES,
UNIVERSITY COLLEGE LONDON

Estimating UK Retail Car-Park Occupancy from High-Resolution Aerial Imagery: A Domain-Adapted Object-Detection Approach

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August 2026



This dissertation is submitted as part requirement for the MSc Emerging Digital Technologies degree at University College London. It is substantially the result of my own work except where explicitly indicated in the text. The dissertation may be freely copied and distributed provided the source is explicitly acknowledged.

Abstract

Car-park occupancy is a valuable but hard-to-measure proxy for how busy a retail site is. Aerial imagery could measure it at scale, yet three obstacles stand in the way. Detecting cars from overhead is a difficult “small-object” problem, no public labelled dataset covers United Kingdom retail car parks, and detectors trained on foreign imagery cannot be assumed to transfer to them. This dissertation builds a deep-learning pipeline to overcome these obstacles. An object detector is first trained on two public overhead-vehicle datasets matched to the roughly 25 cm-per-pixel resolution used here, then adapted on a purpose-built set of thirty hand-labelled UK car parks. Counting accuracy is assessed separately from detection accuracy throughout. Adapting on just eight UK sites raised detection accuracy to 0.886 mean average precision and recovered occupancy to within 2.6 percentage points of capacity on unseen sites, whereas unadapted detectors could not count reliably. Since each detected car carries a location, the counts are read spatially as well as totalled. Across the test sites, parking concentrates around the store’s pedestrian entrance rather than the vehicle entrance, adding a within-lot picture of where demand falls that a single occupancy figure hides. The work shows that a small, in-domain calibration set contributes more than a large volume of out-of-domain data, that a strong detection score does not guarantee an accurate count, and that imagery at around 25 cm resolution suffices for a task on which the coarser satellite imagery of earlier UK work did not.

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List of Abbreviations

AMP	Automatic Mixed Precision
AOI	Area of Interest
AP	Average Precision
BNG	British National Grid
CARPK	Car Parking Lot (vehicle-counting dataset)
CNN	Convolutional Neural Network
COCO	Common Objects in Context (dataset)
COWC	Cars Overhead With Context (dataset)
CVAT	Computer Vision Annotation Tool
DOTA	Dataset for Object deTection in Aerial images
EPSG:27700	British National Grid coordinate reference system
FN	False Negative
FP	False Positive
GDAL	Geospatial Data Abstraction Library
GIS	Geographic Information System
GSD	Ground Sampling Distance
HOG	Histogram of Oriented Gradients
IoU	Intersection over Union
ITCVD	ITC Vehicle Detection (dataset)
LBP	Local Binary Patterns
MAE	Mean Absolute Error
mAP	mean Average Precision
MAPE	Mean Absolute Percentage Error
NMS	Non-Maximum Suppression
OBB	Oriented Bounding Box
R-CNN	Region-based Convolutional Neural Network
RMSE	Root Mean Square Error
RPN	Region Proposal Network
SAHI	Slicing-Aided Hyper Inference
SGD	Stochastic Gradient Descent
SOD-YOLO	Small-Object-Detection YOLO
SSD	Single Shot MultiBox Detector
SVM	Support Vector Machine
TP	True Positive
VEDAI	Vehicle Detection in Aerial Imagery (dataset)
YOLO	You Only Look Once

Acknowledgements

I would like to express my gratitude to my industry and academic supervisors, Toby Merritt and Dr David Vidal Tomas respectively, for their guidance and support throughout this project. I am especially grateful to Knight Frank for giving me the opportunity to work with them and for their practical help along the way.

1 Introduction

How busy is a shop, and could we tell from space? For anyone who manages retail property the answer carries real commercial value, yet it is surprisingly hard to obtain. Direct measures of in-store activity are costly to collect and rarely comparable from one site to the next. For this reason car-park occupancy has become a widely used stand-in for retail trade (Leblanc et al. 2021, Mundhenk et al. 2016, Katona et al. 2025). The logic is simple. For most out-of-town and edge-of-centre stores the car park is the main gateway for customers, so watching it fill and empty over time gives an indirect but informative signal of trade.

The commercial stakes reach well beyond the shop floor. Retail property is a major asset class, and the value of a store, a retail park or an entire portfolio ultimately rests on the flow of customers it attracts. Landlords set rents, investors price assets, and retailers decide where to open or close on the strength of that footfall. Yet such activity is difficult to observe consistently, and harder still to compare across owners. A retailer knows its own sales but not those of its competitors, and an investor weighing a portfolio has no direct way to set a store in one town against a store in another. This information gap has fuelled rapid growth in so-called *alternative data*, where non-traditional signals are used to gauge economic activity that conventional statistics miss or report only with a long delay (Zhu 2019, Katona et al. 2025). An independent, even-handed read on how a site is performing, repeatable across many locations, is precisely the kind of signal this market prizes.

The established ways of gauging that activity each fall short in a different way. Dedicated footfall counters, whether infrared beams, cameras or Wi-Fi and Bluetooth sensors, are accurate at a single door but must be installed and maintained site by site, which makes portfolio-wide coverage slow and costly (Soundararaj et al. 2020). Mobile-location data promises broader reach, yet its coverage is uneven, its sampling is biased toward particular groups of users, and its representativeness is often poorly documented (González et al. 2008, Lu et al. 2017). Transaction and loyalty-card records capture only a retailer’s own customers and are rarely shared. None of these sources gives an outside analyst a uniform read across many competing sites at once, which is exactly the advantage point that an observation from above can provide.

Aerial and satellite imagery offers a way to scale that signal from one site to many (Katona et al. 2025). Rather than fitting equipment at every location, a single system can observe hundreds of car parks from above. The difficulty is that turning such imagery into reliable counts is a hard computer-vision problem. Seen from directly overhead a car spans only a handful of pixels, has no height to give it away, and shows almost no distinguishing detail. Detecting it is therefore a *small-object* problem in the technical sense (Razakarivony &

Jurie 2016, Mundhenk et al. 2016). At this scale the resolution of the image (how much ground each pixel covers) largely decides whether a car can be seen at all or simply blurs into the surrounding asphalt (Gliubičiūtė et al. 2023, Shermeyer & Van Etten 2019).

The idea of reading economic activity from overhead is not new. Night-time lights have long been used as a proxy for economic output and population (Henderson et al. 2012), and counting cars in retailers’ car parks from satellite imagery has become a recognised technique for anticipating sales and company performance (Zhu 2019, Feng & Fay 2022). What has changed recently is that two enabling conditions have matured at once. High-resolution overhead imagery is now far more widely available, and deep-learning object detectors have advanced to the point where they can, in principle, pick out objects only a few pixels across. Together these developments make counting individual vehicles across many sites a realistic proposition rather than a manual, one-off exercise. The question this project addresses is whether that promise holds specifically for dense United Kingdom retail car parks, which differ markedly from the sparser, often American scenes on which most public methods have been developed and tested.

Taken together, these obstacles expose a clear gap in existing UK work. The closest UK study never produced a system that worked in practice, mainly because it relied on coarse satellite imagery of roughly 0.6–1.0 m per pixel, well below the resolution the task demands (Leblanc et al. 2021). It also merged detection and counting into a single step, so it could not distinguish a model that was missing cars from one that was miscounting the cars it found. Compounding both problems, no public labelled dataset covers UK retail car parks. This leaves a domain-shift problem, in which a detector trained on sparse or non-UK imagery cannot be assumed to transfer to the dense, tightly packed and frequently occluded car parks common in the UK (Weiss et al. 2016).

This project confronts those problems with a deep-learning pipeline built on the YOLOv8 detector (Jocher et al. 2023), a modern real-time object-detection model. It sidesteps the resolution limit of earlier UK work by using imagery at about 25 cm per pixel, which sits inside the range shown to give good accuracy for this task (Gliubičiūtė et al. 2023), and trains the detector by transfer learning (Yosinski et al. 2014) from public overhead-vehicle datasets that are resampled to match that resolution before the model is adapted to, and tested on, hand-labelled UK data. The central argument of this dissertation follows from that design. The barrier to measuring UK retail activity from above has been one of *resolution and evaluation* rather than of method. Given imagery at a sufficient resolution and a detector adapted to the target domain, vehicles in dense UK car parks can be counted accurately enough to serve as an occupancy signal, provided that detection and counting are measured *separately* rather than conflated, so that an accurate count is never mistakenly credited to a model whose detection errors merely cancel out.

A secondary contribution follows from the same output. Because each counted vehicle is individually located, the occupancy estimate need not stop at a single per-site total: the detections can be mapped, revealing where within a lot demand concentrates, a spatial dimension the aggregate count discards.

1.1 Research objectives

The main aim of this dissertation is to build and evaluate a pipeline that detects and counts vehicles in dense UK retail car parks from roughly 25 cm aerial imagery, and in doing so to lay the groundwork for using those counts as an occupancy-based measure of retail activity. A further aim is to characterise where within a lot that demand concentrates. Six objectives follow from this aim:

1. **Set the architecture and resolution baseline.** Review the computer-vision and remote-sensing literature to choose a suitable detector and to find the coarsest resolution that still supports small-object vehicle detection from above.
2. **Build resolution-matched training data.** Prepare and standardise training data from two public overhead-vehicle datasets: Cars Overhead With Context (COWC) (Mundhenk et al. 2016) and Vehicle Detection in Aerial Imagery (VEDAI) (Razakarivony & Jurie 2016).
3. **Create a UK test set.** Hand-label a UK retail car-park dataset from Aerial Digimap (Getmapping) imagery (Getmapping Plc 2021), to act as a genuine, held-out test in the target domain.
4. **Measure what source training and UK adaptation each add.** Using a controlled, three-part comparison on a frozen UK test set, separate the accuracy that comes from generic pretraining alone (baseline), from transferring the public models to the UK with no UK training (transfer), and from fine-tuning on UK data (UK-adaptation).
5. **Estimate and evaluate occupancy.** Turn vehicle detections into site-level counts and occupancy rates, and evaluate them against hand-annotated ground truth, reporting detection quality and counting error separately.
6. **Map the geography of demand.** Re-express the located detections as georeferenced spatial data and test how within-lot demand relates to a site’s points of access, quantifying whether parking concentrates near the retail (pedestrian) or the vehicle entrance.

1.2 Research methodology

The project uses an *inductive, homogeneous* transfer-learning approach (Weiss et al. 2016, Pan & Yang 2010) to work around the shortage of labelled UK retail imagery. Source and

target share the same kind of input (three-channel RGB overhead imagery) and the same task (detecting a single *vehicle* class), so what the model learns from the public data can carry over to the UK, making up for the limited UK training data. Three data sources are used: the dense, urban scenes of COWC (Mundhenk et al. 2016); the sparser, more varied scenes of VEDAI (Razakarivony & Jurie 2016); and a new UK retail set obtained through UCL’s Digimap licence (Getmapping Plc 2021).

The preparation pipeline runs in clear steps. The public source scenes are downsampled to the shared 25 cm resolution so that all training data lines up geometrically; centre-point labels are converted into fixed-size square boxes; and the large scenes are cut into 640×640 tiles using an overlapping sliding window that avoids losing objects at tile edges (Akyon et al. 2022). Splits are made at the scene (or site) level, not the tile level, so that overlapping tiles cannot leak the same cars across the train/test boundary.

Evaluation happens in two stages. First, the single-source models (COWC, VEDAI) and a combined model are compared within and across the two public domains, which measures the gap between them. Second, and this is where the main findings lie, three detectors of increasing adaptation are tested on the *same* held-out UK sites: a stock baseline, the combined model applied with no UK training, and that model after fine-tuning on UK data. Throughout, detection metrics (mAP@0.5 and mAP@0.5:0.95) are kept separate from counting accuracy, so that a localisation problem is never mistaken for a counting problem.

Finally, because every detection carries a location, the site-level counts are re-expressed as georeferenced points, and the within-lot distribution of demand is analysed against hand-digitised store and car-park entrances to establish how parking is organised relative to a site’s access points.

1.3 Dissertation structure

The rest of the dissertation is organised as follows.

Section 2 (*Literature Review*) sets out the core concepts, frames vehicle detection from above as a small-object problem, examines how much image resolution matters and why dense, occluded scenes are hard, explains the choice of a one-stage YOLOv8 detector, introduces the transfer-learning framework, and pins down the gap this project fills.

Section 3 (*Methodology*) describes the data-preparation pipeline of box conversion, resolution matching, tiling and scene-level splitting, together with the YOLOv8 training setup, the UK dataset, the design of the three UK experiments, and the counting-and-occupancy protocol built on top of the adapted detector.

Section 4 (*Results*) reports the findings, covering how the source models behave within

and across domains, how the three UK detectors compare on precision, recall, F1 and mAP, how accurately the adapted detector translates into vehicle counts and occupancy estimates, and, treating each detection as a located point, how within-lot demand is distributed relative to a site's entrances.

Section 5 (*Discussion*) interprets the domain gaps, examines the limits of axis-aligned square boxes in dense car parks, assesses how workable the occupancy estimate proves to be, and interprets the within-lot geography of demand, in particular its organisation around the store entrance.

Section 6 (*Conclusion*) draws the contributions together, notes what they offer retail and real-estate analysis, and sets out future work.

2 Literature Review

2.1 Conceptual foundations

Object detection is the computer-vision task of locating instances of an object within an image and marking each with a bounding box. It is more demanding than image classification, which asks only whether an object is present, and less demanding than instance segmentation, which recovers the exact pixel outline of each object (Zou et al. 2023) (Figure 1). The present work uses detection: a bounding box is drawn around each vehicle, which is sufficient to count cars and to estimate occupancy without the additional cost of pixel-level labelling.

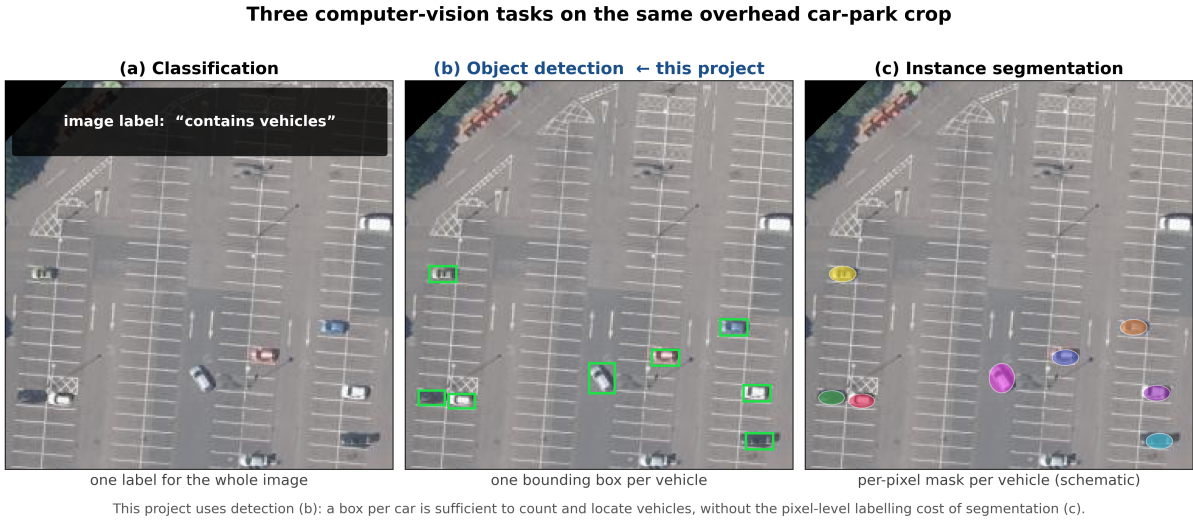


Figure 1: Computer-vision tasks illustrated on a 25 cm UK retail car-park crop: (a) classification (single global label); (b) object detection (bounding boxes used in this project); and (c) instance segmentation (per-pixel masks, shown here for illustration only).

Modern detectors are built on convolutional neural networks (CNNs), which learn visual features directly from image data through a hierarchy of layers, progressing from simple edges to whole-object structure (LeCun et al. 1998, 2015). Two properties of CNNs are particularly relevant here. First, weight sharing means a feature learned in one part of the image can be detected anywhere in it, so a vehicle in one corner of an aerial scene is recognised as readily as one in another (LeCun et al. 1998). Second, a network pretrained on a large generic dataset can be adapted to a narrower task with comparatively little data through transfer learning, which is the approach taken in this dissertation and is examined in detail in Section 2.6. Since the breakthrough of deep CNNs on large-scale image classification (Krizhevsky et al. 2012), detectors have typically been built on progressively deeper classification backbones such as the residual networks that made very deep models trainable (He et al. 2016), and it is these backbones that a detector reuses when it is fine-tuned by transfer learning.

Detector performance is assessed using a small set of standard measures (Padilla et al.

2020). Intersection over Union (IoU) quantifies how well a predicted box aligns with the ground-truth box, by dividing the area of their overlap by the area of their union, giving a value between 0 (no overlap) and 1 (perfect alignment). A detection is typically accepted as correct when its IoU exceeds a threshold such as 0.5 (Everingham et al. 2010). Mean Average Precision (mAP) condenses a detector’s overall performance into a single figure by taking the area under the precision–recall curve for each class and averaging across classes, thereby capturing detection quality and localisation quality together (Padilla et al. 2020). Two conventions are reported throughout this dissertation: mAP@0.5, which counts a detection as correct at an IoU of 0.5 and the stricter mAP@0.5:0.95, averaged over IoU thresholds from 0.5 to 0.95, which is the COCO standard and penalises loose localisation (Lin et al. 2014). Since the avoidance of duplicate detections is crucial to accuracy, Non-Maximum Suppression (NMS) is applied to remove overlapping boxes that describe the same object (Neubeck & Van Gool 2006).

2.2 Vehicle detection from aerial imagery as a small-object problem

A consistent finding across the aerial-vehicle detection literature (Li et al. 2020) is that detecting cars from above is not ordinary object detection but a small-object problem. Each target occupies only a handful of pixels and carries little distinguishing information. Razakarivony & Jurie (2016) make this explicit in introducing VEDAI, a benchmark for small-vehicle detection in aerial imagery, in which vehicles occupy roughly 0.7% of the image pixels and there are on average only about five-and-a-half vehicles per frame.

They highlight the severe limits of pre-deep-learning detectors. Traditional methods based on hand-crafted features, such as Support Vector Machines (Cortes & Vapnik 1995) using HOG (Dalal & Triggs 2005) or LBP (Ojala et al. 2002) descriptors, along with deformable part models (Felzenszwalb et al. 2010) and template matching, struggle significantly. In fact, the top-performing car detector recovered only about 13% of vehicles at a realistic operating point. Ultimately, the authors conclude that none of the conventional detectors available at the time offered usable performance.

This verdict must be read carefully. It concerns 2015-era classical methods, not the convolutional detectors that have since become standard, and it should therefore be treated as motivation for the deep-learning approach adopted here rather than as a statement about the current limits of the field. Framed this way, VEDAI’s contribution is to establish that vehicle-from-above is intrinsically a small-target task, justifying a dedicated thread of literature on small-object difficulty (Tong et al. 2020) and supporting design choices such as a high input resolution and resolution-matched training data.

The same characterisation recurs in the dataset used as the primary training source in this project. Mundhenk et al. (2016), introducing the Cars Overhead With Context (COWC) dataset, standardise their imagery to 15 cm per pixel, at which a car spans only 24 to

48 pixels. The persistence of this framing from the classical work of 2015 through to the convolutional work of 2016 and beyond indicates that spatial detail, and therefore image resolution, is the decisive variable for this task.

2.3 The influence of image resolution on detectability

If vehicle detection from above is fundamentally limited by how few pixels each car occupies, then the resolution of the source imagery should be the single most important determinant of accuracy. Gliubičiūtė et al. (2023) test this directly and, in doing so, provide the closest methodological precedent to the present work. They use both datasets adopted here (COWC and VEDAI), the same model family (YOLOv8), and the same method (transfer learning from pretrained weights), to answer precisely the question that motivates this project: how far imagery can be degraded before detection accuracy collapses. They reduce both datasets to a single car class and progressively downsample them to 15, 20, 25, 27.5, 30 and 32.5 cm per pixel, fine-tuning YOLOv5, YOLOv7 and YOLOv8 at each step.

Their results convert a qualitative intuition into a citable curve. Accuracy degrades only gently across the band of interest, with coarsening in 5 cm steps from 12.5 to 27.5 cm per pixel costing on average about 3.5% mAP across all models. YOLOv8 is the most resolution-robust of the three, and at 25 cm per pixel (the resolution of the UK imagery used in this dissertation) it achieves an mAP@0.5 of approximately 0.93 on COWC and 0.92 on VEDAI, barely below its native-resolution score. Degradation accelerates sharply only beyond 30 cm (Figure 2). Two implications follow directly. Firstly, the choice of 25 cm imagery sits firmly within the stable, high-accuracy region rather than representing a compromise. Secondly, the finding is a direct rebuttal to the principal failure in the prior UK work discussed in Section 2.8: Leblanc et al. (2021), working with 0.6 to 1.0 m satellite imagery, were operating well past the accuracy cliff, which provides strong evidence for the change of imagery at the centre of this project.

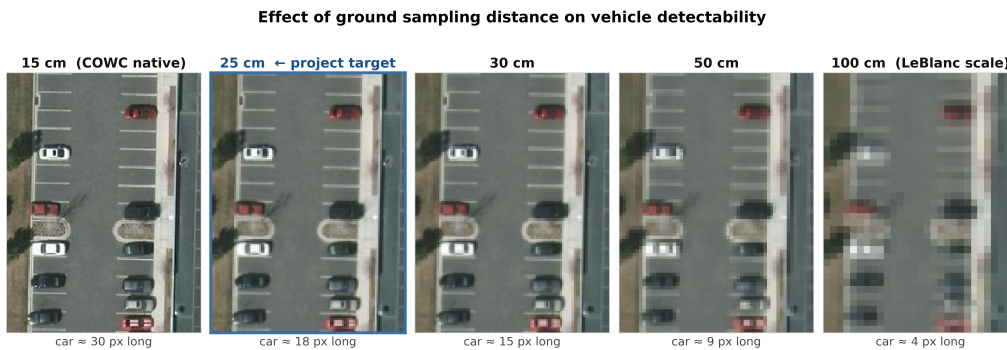


Figure 2: Effect of ground sampling distance on vehicle detectability, simulated from 15 cm COWC imagery using bilinear downsampling. Features remain stable within the 25–30 cm target band (highlighted), but collapse to approximately 4 px in length at a 100 cm GSD, representing the metre-scale satellite imagery regime of Leblanc et al. (2021).

Several limitations affect how these figures are used. The headline scores of around 0.92 to 0.93 are for single-class car detection on relatively sparse scenes and are therefore not comparable to the lower figures reported for dense, multi-class benchmarks discussed later. It is the shape of the accuracy-against-resolution curve, not the absolute values, that constitutes the citable contribution. The pronounced collapse the authors report for VEDAI at low resolution is also likely to be in part an artefact of that dataset’s small size (around 2,800 annotated boxes) and its labelling of partially hidden and edge-clipped vehicles, rather than a pure effect of resolution. The larger COWC curve is therefore the more trustworthy guide for the dense retail scenes targeted here. The study’s reporting is moreover internally inconsistent in places, and it rests on single runs on a single GPU with a small test split, so it is best treated as indicative.

With these qualifications, Gliubičiūtė et al. (2023) remains the strongest single source for the fine-grained *shape* of the accuracy–resolution curve. The broader conclusion, that approximately 25 cm aerial imagery is sufficient for car detection while coarse satellite imagery is not, is corroborated independently by Shermeyer & Van Etten (2019), whose satellite study reports detection accuracy falling from an mAP of 0.53 at 30 cm to 0.11 at 4.8 m, with the greatest gains coming from the finest resolutions.

2.4 Scene density, clustering and occlusion

Retail car parks present a second difficulty distinct from small object size. Vehicles are densely packed, frequently touching, or partially occluding one another. The literature acknowledges this difficulty consistently (Xia et al. 2018) but treats it very differently from dataset to dataset, which has direct consequences for which sources are suitable as training data. Leblanc et al. (2021) attribute much of their under-detection to clustered and occluded cars. Razakarivony & Jurie (2016), by contrast, deliberately exclude images containing many vehicles, such as large car parks, from VEDAI, on the grounds that an algorithm guessing random positions would score deceptively well on them. The consequence is that VEDAI scenes are sparse and, as the authors note, NMS rarely matters. This makes VEDAI’s scene statistics sharply different from those of the dense retail car parks of interest here, even though it remains valuable for teaching a model what an overhead car looks like.

Mundhenk et al. (2016) take the opposite approach, COWC explicitly includes dense parking scenes. The authors observe that image patches in crowded lots overlap and that occlusion and the merging of adjacent vehicles are the hard cases for counting. COWC therefore covers precisely the dense, occluded condition that VEDAI excludes and that defeated the coarse-resolution work of Leblanc et al. (2021) (Figure 3). This contrast is the central reason COWC is adopted as the primary training source in this project, since it carries the dense-scene training load that VEDAI cannot. The UK panel shows the *target*

domain that motivates this choice, the UK dataset itself is constructed in Section 3.7. It also reinforces why neither source removes the need for domain-specific UK validation data, since neither was captured in UK retail settings.

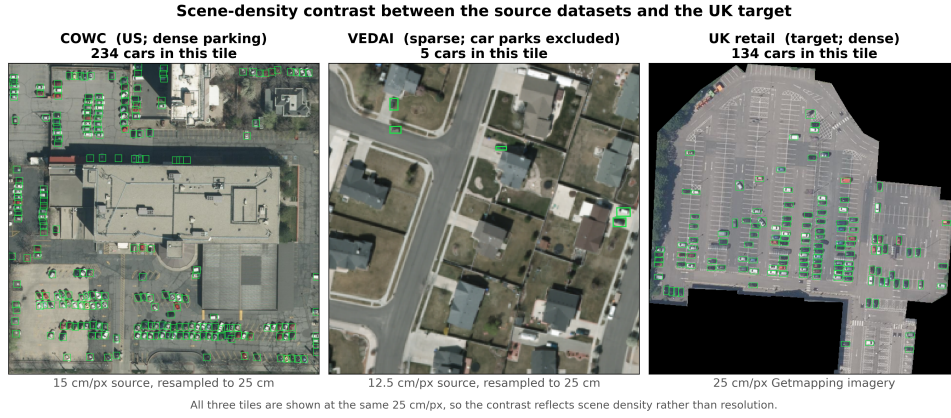


Figure 3: Scene-density comparison across the public source datasets and the UK target. COWC’s dense parking layout closely mirrors the highly clustered conditions of the UK retail target, whereas VEDAI deliberately omits large parking lots, resulting in a highly sparse distribution. Green boxes represent ground-truth vehicle annotations.

Two further aerial benchmarks sit close to this task and are worth positioning against the two sources adopted here. ITCVD (Yang et al. 2019) provides airborne imagery of a European city at roughly 10 cm per pixel with tight, individually drawn vehicle boxes. In annotation geometry it sits closer to the UK target than either public source used in this project, since its boxes match the tight axis-aligned form of the UK ground truth rather than the fixed-size squares COWC’s centre points impose. DOTA (Xia et al. 2018), by contrast, is a large multi-class aerial benchmark of some 400,000 instances annotated with *oriented* bounding boxes, which makes it the natural training source for the rotated-box detection considered as future work (Section 5.1). Both are therefore relevant to the design choices of this project; the practical reasons neither was adopted as a training source are set out in Section 5.5.

2.5 Detector architectures and the choice of YOLOv8

Deep object detectors fall into two broad families, and the choice between them is a recurring architectural fork in the literature. Two-stage detectors first propose candidate regions and then classify and refine them. On the other hand, one-stage detectors predict boxes and classes directly in a single pass, trading some accuracy for considerably greater speed. The two-stage line began with R-CNN (Girshick et al. 2014) and culminated in Faster R-CNN (Ren et al. 2015), which introduced a Region Proposal Network to generate candidates almost for free and, in the process, introduced the anchor boxes that most subsequent detectors inherited. Anchor boxes are a fixed set of reference boxes of pre-chosen sizes and shapes tiled across the image, so that the detector predicts small adjustments to the nearest reference rather than a box from scratch. The relevance of

anchors to small-object detection is shown by the authors' own decision to add a smaller anchor scale specifically to handle small objects on the COCO dataset, demonstrating that default anchor priors sized for everyday objects must be shrunk for tiny targets.

Faster R-CNN is best understood here as the alternative this project rejects rather than as a candidate method. Although two-stage detectors have historically been more accurate on generic objects, they are consistently weaker on small objects and slower. In the comparison reported by Li et al. (2024), Faster R-CNN reaches only 37.2 mAP@0.5 on the aerial VisDrone benchmark (Zhu et al. 2021), well below one-stage YOLO variants, because the proposal network tends to generate few or poor proposals at small scales. For a small-target, near-real-time aerial task, a one-stage detector is therefore the better-motivated choice.

Within the one-stage family, the principal alternative to YOLO is the Single Shot MultiBox Detector (Liu et al. 2016). However, SSD remains anchor-based and detects small objects from relatively shallow feature maps. A design its own authors identify as the source of its comparatively weak small-object performance, and a key reason later detectors adopted multi-scale feature pyramids (Lin et al. 2017). By contrast, the YOLO lineage has evolved rapidly (Murat & Kiran 2025). The series moved from the original anchor-free design (Redmon et al. 2016), through anchor-based versions (Redmon & Farhadi 2018) that added feature pyramids, to a recent return to anchor-free architectures that predict object centers directly (Tian et al. 2019).

YOLOv8 (Jocher et al. 2023) exemplifies this modern direction. It operates anchor-free and uses a decoupled detection head to separate classification from box regression. An arrangement popularized for the series by Ge et al. (2021). By predicting object centers directly rather than regressing offsets from preset anchors, it eliminates manual anchor tuning. This in turn yields fewer candidate boxes and faster non-maximum suppression (Figure 4). This design offers a distinct advantage for the present task: it avoids the headache of matching preset anchor sizes to tiny aerial vehicles entirely by construction. Combined with mature tooling and the fact that the most directly comparable resolution study (Gliubičiūtė et al. 2023) also uses YOLOv8, this makes it a solid, defensible baseline. (Note that while Murat & Kiran (2025) provides a helpful overview of this lineage, their review contains consolidated performance tables that mix datasets and input sizes, along with at least one likely misclassification of anchor status; it is cited here solely for historical context, with all technical details drawn from primary sources.)

This choice is further reinforced by direct, application-level evidence. Shreeram H et al. (2025) benchmark a range of YOLO variants for parking-space detection in aerial imagery and identify YOLOv8 and YOLOv9 as the two top-performing architectures. YOLOv9 achieved the highest overall accuracy, but at a noticeably higher computational cost,

whereas YOLOv8 offered the best balance between speed and accuracy.

For a pipeline designed to scale across numerous commercial sites on modest hardware, that balance is the deciding factor. Consequently, YOLOv8 is adopted here over the marginally more accurate but resource-intensive YOLOv9 (Wang & Liao 2024). Notably, this trade-off is domain-specific: while YOLOv9 is reported as the more parameter-efficient architecture on the general MS COCO benchmark, the empirical evidence on aerial parking imagery favors YOLOv8 for scalable deployment.

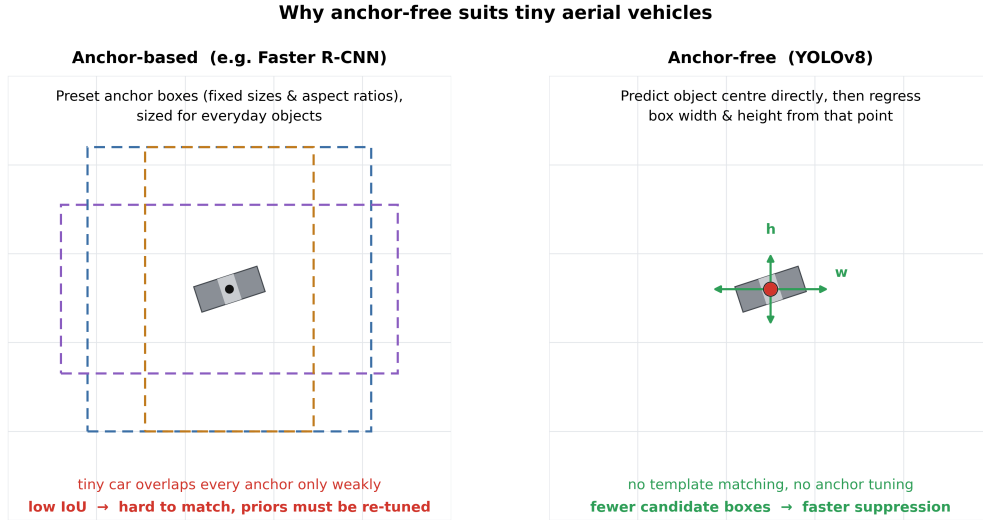


Figure 4: Comparison of detection paradigms on a small aerial vehicle. **Left:** Anchor-based methods rely on fixed templates, yielding low IoU for tiny targets and requiring manual prior tuning. **Right:** Anchor-free methods (e.g., YOLOv8) directly predict center points and dimensions, bypassing template matching.

The most directly relevant modern evidence comes from Li et al. (2024), whose SOD-YOLO modifies YOLOv8 specifically for small targets in aerial imagery. Their central finding is that stock YOLOv8 struggles with tiny targets, and that the fix is architectural. Specifically, incorporating a higher-resolution detection path and stronger multi-scale feature fusion, rather than simply adding more data. These modifications boost small-object AP by 6.4 points over the standard baseline. This offers strong, contemporary evidence that detecting vehicles from above remains a distinct small-object challenge, while also reaffirming YOLOv8 as the right baseline to build on. However, two important qualifications apply. First, absolute accuracy on this type of data remains low across the board, peaking at only around 42 to 51 mAP@0.5 on VisDrone. This context is helpful for setting realistic expectations, ensuring a modest mAP score on small cars is recognized as typical rather than a model failure. Second, and more critically, there is a significant domain gap that runs counter to this project. VisDrone consists of low-altitude, multi-angle street footage, whereas the imagery used here comprises high-altitude, fixed-nadir orthophotos of car parks. Since SOD-YOLO’s performance gains are explicitly

designed to handle viewpoint and scale variation, they cannot be assumed to transfer to a scale-consistent top-down dataset. Consequently, SOD-YOLO serves as a well-supported backup option if fine-tuning stock YOLOv8 falls short, rather than the primary starting architecture.

2.6 Domain shift and transfer learning

Every dataset reviewed above was captured outside the United Kingdom and at a resolution differing from the target imagery. Consequently, the generalization of any detector trained on them to UK retail car parks cannot be assumed. The survey by Weiss et al. (2016) supplies the formal framework and vocabulary to state this problem precisely. Transfer learning improves a target model by reusing knowledge from a related source task. It is particularly valuable when target data is scarce or expensive to label, exactly the situation here, given the limited availability of UK retail aerial imagery. Formally, a domain consists of a feature space paired with a data distribution, while a task consists of a label set paired with a predictive function. Transfer learning applies whenever the source and target differ in either element.

Using this framework, the present project is characterized specifically as *inductive, homogeneous* transfer learning subject to a marginal-distribution shift. The setting is homogeneous because the source and target share both a feature space (RGB aerial imagery) and a task (detecting cars). It is inductive because labeled target data is actively produced. The domain gap manifests as a marginal-distribution shift: the input distribution varies between source and target due to differences in vehicle appearance, parking layouts, camera sensors, and spatial resolutions, even though the label space remains identical. This is precisely the scenario addressed by deep visual domain adaptation (Wang & Deng 2018). It recurs frequently in remote sensing, where imagery captured across different sensors, seasons, or regions shifts the input distribution while leaving the core task unchanged (Tuia et al. 2016). Framing the problem this way elevates an informal "domain gap" into citable theory that underpins the entire review.

The survey's treatment of negative transfer, where an insufficiently related source degrades target performance, provides the most actionable insight. Considering that the risk of negative transfer declines as labeled target data increases and source-target similarity rises, this framework provides theoretical backing for three key project decisions:

1. Annotating a custom UK dataset to enable inductive fine-tuning and mitigate negative transfer risk;
2. Selecting overhead vehicle datasets and aligning their spatial resolution with the target to maximize domain similarity; and

3. Treating the downsampling of COWC’s 15 cm imagery to 25 cm as a lightweight domain adaptation step that narrows the marginal gap.

While Weiss et al. (2016) offers an enduring taxonomy (attributable fundamentally to Pan & Yang (2010)), it predates modern deep learning fine-tuning. Thus, it is cited here for its theoretical framework rather than for specific operational methods. The empirical basis for the fine-tuning approach used in this work rests instead on Yosinski et al. (2014). They demonstrate that features learned on a large source dataset transfer effectively to related target tasks, and that fine-tuning those transferred layers significantly enhances generalization, even under modest domain shifts.

A concrete instance of this shift in the precise setting of the present work is provided by Fang et al. (2025). They show that vehicle detectors trained on aerial imagery from one region lose a substantial share of their AP_{50} when applied to another, and that the gap narrows only once some target-domain information is introduced. Their study confirms both that the marginal-distribution shift anticipated here is real for overhead vehicles specifically, and that a small quantity of in-domain supervision is the effective lever for closing it. Where they synthesise target-like data through generative augmentation, this project supplies that in-domain signal directly, through a small hand-labelled UK calibration set.

2.7 Remote sensing as a measure of economic activity

The preceding sections establish how to build a vehicle detector in overhead imagery, ensure its robustness to coarse resolutions, and adapt it across domains. The review now turns from those technical mechanisms to the applied problem they serve. The core premise of this project, that overhead imagery can measure economic activity otherwise expensive or impractical to observe directly, is well established across both economics and remote sensing.

Indeed, Donaldson & Storeygard (2016) survey a broad and growing body of economics research that uses satellite imagery as a primary data source when conventional statistics are scarce, unreliable, or overly coarse. Their findings place the retail-scale proxy pursued here within an established methodological tradition, rather than treating it as a one-off experiment.

The foundational evidence for this approach originates from nighttime satellite imagery. Henderson et al. (2012) demonstrated that nighttime light intensity closely tracks gross domestic product, providing reliable growth estimates at regional and sub-national scales where official statistics are often weak. This paradigm extends naturally to daytime observation: Jean et al. (2016) combined high-resolution daytime satellite images with transfer learning to predict local economic indicators, such as household consumption and

wealth, across multiple African nations. These studies confirm a core principle: visual features extracted from overhead imagery carry a strong, measurable economic signal.

This principle translates directly to the scale of individual commercial sites. Where nighttime lighting proxies aggregate macro-economic output, vehicle counts in a commercial car park proxy the specific economic activity of that store. A fuller lot reflects higher customer footfall, while occupancy fluctuations track commercial performance over time. This concept underpins the growing field of alternative data in financial analytics, where geospatial observations are used to anticipate earnings ahead of official corporate filings (Katona et al. 2025). Car park occupancy is therefore not an arbitrary metric, but a micro-scale application of a proven remote sensing methodology.

The economic premise, however, is not where the difficulty lies. Whether a car park’s occupancy can be read from an image at all depends on the detection problem set out in the sections above: the imagery must first be turned into an accurate count of vehicles, and it is there, rather than in the proxy argument, that the technical challenge sits. The following section examines how prior work has attempted exactly this, counting cars from overhead imagery for the retail-activity purpose pursued here, and where those attempts have fallen short.

2.8 Overhead car counting as a proxy for retail activity

The applied premise of this dissertation, that car-park occupancy can indicate in-store retail intensity, rests on the established track record of overhead imagery as an economic proxy set out in Section 2.7. Two studies anchor that premise most directly for the computer-vision task pursued here. Leblanc et al. (2021), a UCL study conducted with the retailer Sainsbury’s, explicitly frames satellite-derived car-park occupancy as a proxy for in-store trade. It represents the closest UK precedent to the present work in terms of task, framing, model family, and retail context.

Its pipeline, however, severely under-detected vehicles, with detected counts explaining only around 11% of those observed. Beyond the coarse imagery already discussed, its central limitation is that detection and counting are conflated. Only a count-regression statistic is reported, with no precision, recall or mAP, so detection quality cannot be separated from counting quality. Resolving this conflation, by reporting detection metrics and counting error separately, is a specific aim of the present project.

More recent work has moved from coarse satellite imagery to fine-resolution aerial imagery. Hellekes et al. (2023) detect individual vehicles and parking spaces from aerial images and estimate occupancy for unobserved regions, showing that fine-resolution overhead imagery supports space-level parking analysis of a kind the metre-scale satellite studies could not. It does not, however, address the UK retail setting or separate detection from

counting, so the specific combination this project targets remains open.

A second independent anchor is provided by Mundhenk et al. (2016), who introduce the COWC dataset by citing commercial applications that track retail parking lot traffic to forecast business volume for financial investors. Their work also illustrates an alternative methodological paradigm: direct density-based regression, which estimates vehicle counts within an image patch without predicting bounding boxes.

This density-based paradigm originates with Lempitsky & Zisserman (2010), who advocate learning a density surface precisely to avoid committing to individual detections, on the grounds that per-object false positives and false negatives accumulate into the total rather than cancelling. That observation cuts both ways for the present work. It is exactly why a strong detection score cannot be assumed to yield an accurate count, and hence why this project reports detection quality and counting error as separate quantities rather than inferring one from the other.

In contrast, the present project adopts a *detection-based* counting approach, locating each vehicle individually prior to tallying. Explicit detection is necessary here because spatial locations are required for occupancy mapping and qualitative interpretability, a methodology exemplified by the CARPK benchmark (Hsieh et al. 2017). The choice between detection-based and density-based counting is thus a deliberate design decision, directly aligned with the downstream goal of mapping store-level occupancy.

2.9 Synthesis: the gap this project addresses

Taken together, the literature establishes a coherent set of positions. Vehicle detection from above is a small-object problem in which resolution is decisive. Dense, occluded car parks are the hardest and most relevant case, and one-stage anchor-free detectors such as YOLOv8 are the appropriate modern tool. Non-UK source data cannot be assumed to transfer to UK retail settings without domain adaptation and some target labelling. Finally, overhead car counts are an established proxy for retail activity.

Against these positions, no single study occupies the specific niche this project targets (Figure 5). Several come close, but each falls short in a different way. LeBlanc’s UK work is coarse-resolution and non-adapted, and consequently unsuccessful (Leblanc et al. 2021). VEDAI is a sparse, non-UK benchmark that deliberately excludes the dense case (Razakarivony & Jurie 2016), whereas COWC is a dense but US-based dataset evaluated through one-look counting rather than detection (Mundhenk et al. 2016). SOD-YOLO is a modern detector demonstrated on oblique, multi-class drone footage rather than nadir orthophoto (Li et al. 2024), and Gliaubiute is a resolution sweep on non-UK data that stops short of the occupancy-estimation step (Gliubičiūtė et al. 2023). Fang closes the cross-domain gap with synthetic augmentation rather than in-domain labels, and

outside any retail-occupancy setting (Fang et al. 2025). Hellekes is a high-resolution aerial parking-inventory system that reaches space-level occupancy but neither addresses UK retail nor separates detection from counting (Hellekes et al. 2023). None reports detection and counting quality separately, and none carries the resulting counts beyond an aggregate total to the geography of demand within a lot.

Positioning of prior work against the niche this project addresses

	Fine res. (~25 cm)	UK data	Dense / occluded	Nadir top-down	Detection- based	Det. & count separate	Occupancy estimation	Within-lot mapping
LeBlanc et al. (2021)	✗	✓	✓	✓	✓	✗	✓	✗
Hellekes et al. (2023)	✓	✗	✓	✓	✓	✗	✓	~
VEDAI (2015)	✓	✗	✗	✓	✓	✗	✗	✗
COWC (2016)	✓	✗	✓	✓	~	✗	✗	✗
SOD-YOLO (2024)	✓	✗	✓	✗	✓	✗	✗	✗
Gliaubiute (2023)	✓	✗	✗	✓	✓	✗	✗	✗
Fang et al. (2025)	✓	✗	~	✓	✓	✗	✗	✗
This project	✓	✓	✓	✓	✓	✓	✓	✓

✓ addressed ✗ not addressed ~ partial / indirect (assignments follow the synthesis in this chapter)

Figure 5: Comparison of prior studies against the core criteria defining the project scope. Red cells indicate unaddressed criteria, while the symbol ~ represents partial or indirect coverage, synthesized from the literature review in this chapter.

The contribution of this dissertation is to fill that gap. It detects and counts vehicles in dense UK retail car parks from approximately 25 cm fixed-nadir aerial imagery, using a YOLOv8 detector domain-adapted to that resolution and validated against UK ground truth. It translates those detections into car-park occupancy estimates, with detection and counting quality reported separately. And because each detection is located, it maps how demand is distributed within each lot rather than stopping at an aggregate count. No prior study spans this combination. The chapter that follows sets out the methodology by which it is achieved.

3 Methodology

3.1 Overview of the approach

To detect and count vehicles in UK retail car parks from 25 cm aerial imagery, this project fine-tunes a pretrained YOLOv8 detector on public overhead-vehicle datasets that have been resolution-matched to the target imagery. This executes the inductive, homogeneous transfer learning strategy detailed in Section 2.6: rather than training from scratch, the model is initialized with weights pretrained on a large generic dataset and adapted to the specific domain of aerial vehicle detection (Yosinski et al. 2014).

Data-preparation and training pipeline (applied to each source dataset)

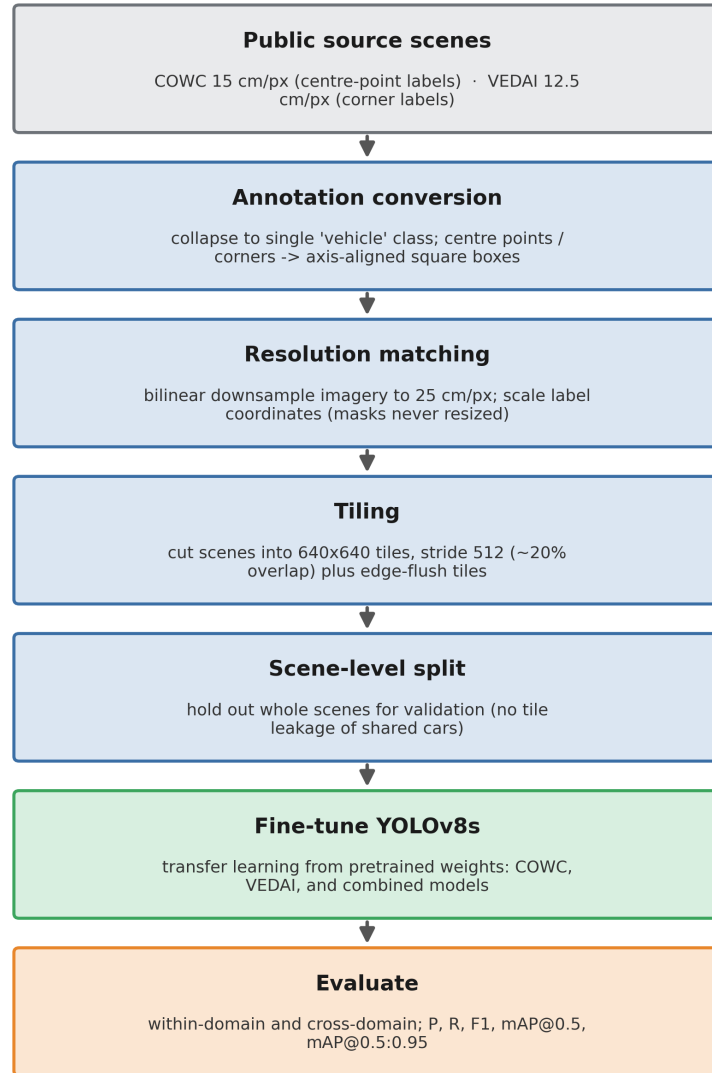


Figure 6: Data preparation and YOLOv8s training pipeline applied to each source dataset. Following class consolidation and resolution-matching to 25 cm/px, imagery is tiled with overlap and split at the scene level to prevent spatial leakage. The validation workflow for the UK dataset follows an identical procedure (Section 3.7).

The overall pipeline proceeds in four main stages (Figure 6):

1. **Annotation Standardization:** Vehicle labels are consolidated into a single target class using axis-aligned bounding boxes.
2. **Resolution Matching & Tiling:** Source imagery is resampled to a spatial resolution of 25 cm per pixel and tiled into uniform, overlapping patches suitable for model input.
3. **Scene-Level Splitting:** Tiled images are partitioned into training and validation sets strictly at the scene level to prevent spatial data leakage across splits.
4. **Model Fine-Tuning:** A pretrained YOLOv8s model is fine-tuned on the processed tiles and evaluated on holdout validation data.

This was carried out independently on two public datasets and on their union, and the resulting models were compared both within and across domains. The final stage of the method, constructing the labelled UK dataset and using the detector to estimate car-park occupancy, is described in Sections 3.7 and 3.10.

For reproducibility, the preparation pipeline was implemented as small, individually tested Python components driven by configuration files, with the detector built on the Ultralytics implementation of YOLOv8 (Jocher et al. 2023). Representative code is provided in Appendix A.

3.2 Datasets

Two public datasets were used. The primary training source was the Cars Overhead With Context (COWC) dataset (Mundhenk et al. 2016), chosen because, unlike most aerial-vehicle benchmarks, it includes dense parking scenes and therefore covers the tightly clustered condition that characterises retail car parks (Section 2.4; see also Figure 3). As inventoried for this project, COWC comprises 32 scenes across six geographically distinct regions, totalling 37,721 annotated cars and 69,551 hard negatives at a standardised resolution of 15 cm per pixel; its composition is summarised in Table 1. A distinctive feature of COWC, which drives the annotation step below, is that vehicles are marked by single centre-point dots rather than bounding boxes.

A second dataset, VEDAI (Razakarivony & Jurie 2016), was used to provide an independent benchmark and to test cross-domain generalisation. VEDAI supplies corner-coordinate annotations across several vehicle classes at a native resolution of 12.5 cm per pixel. In contrast to COWC, it deliberately excludes dense car parks, so its scenes are sparse. Keeping it entirely separate from COWC allowed the two to be compared directly.

The ultimate deployment domain is UK retail car-park imagery at approximately 25 cm per pixel, obtained from the EDINA Aerial Digimap service (Getmapping) (Getmapping

Plc 2021). As established in Section 2.6, neither public source is UK imagery, so a domain gap to the target remains; the labelled UK set used to measure and close this gap is the subject of Section 3.7.

Table 1: Composition of the COWC dataset as inventoried for this project.

Region	Scenes	Cars	Notes
Utah (AGRC)	9	19,807	Densest; a single scene holds 13,213 cars
Toronto (ISPRS)	3	10,023	Dense urban; source of the held-out validation scene
Vaihingen (ISPRS)	1	2,863	Grayscale
Potsdam (ISPRS)	13	2,083	Many small, sparse scenes
Columbus (CSUAV/AFRL)	3	1,748	RGB
Selwyn (LINZ)	3	1,197	Very large, sparse tiles
Total	32	37,721	plus 69,551 hard negatives

3.3 Data preparation pipeline

All source data was standardised to a common form before training: a single vehicle class, axis-aligned bounding boxes, a uniform 25 cm resolution and 640-pixel tile size. The steps were applied to each dataset, with documented differences for VEDAI.

3.3.1 Annotation conversion: centre points to bounding boxes

Since COWC marks each vehicle with a centre-point dot rather than a box, a fixed-size square bounding box was generated around every point (Figure 8). The box size was set from physical geometry and then confirmed empirically. A car of roughly 4.5×1.8 m spans about 18×7 pixels at 25 cm. Squares of 12, 13, 14 and 16 pixels were drawn over the same densely parked cluster (Figure 7), and 14 pixels proved the largest side that still encloses each vehicle without its box merging too much into a neighbour’s. As can be observed, at 16 pixels the boxes began to swallow adjacent cars, while 12 and 13 boxed them loosely. A 14×14 -pixel square (about 0.022 of the tile width when normalised) was therefore adopted.

A square, rather than a fixed-orientation rectangle, was chosen deliberately because COWC provides no orientation information and vehicles appear at all rotations. So an axis-aligned rectangle would fit only one heading and mis-fit all others, whereas a square is rotation-invariant and encloses the vehicle equally at any heading.



Figure 7: Selection of bounding box dimensions on a dense COWC cluster (57 vehicles). Candidate side lengths (12–16 px) were evaluated on identical centroids; the 14×14 px footprint was selected to maximize vehicle coverage while preventing spatial overlap.

The 14-pixel side captures the vehicle’s body consistently across orientations while deliberately staying below the car’s full length of about 18 pixels, trading a little under-coverage of the major axis for tighter boxes that overlap less between densely packed vehicles. This choice, together with the residual over-coverage on the short axis, is the principal reason loose localisation is anticipated in the results, and it motivates the oriented-bounding-box experiment proposed as future work. Box coordinates were normalised against the tile rather than the full scene, and box corners falling outside a tile were clipped to its boundary (Listing 1).

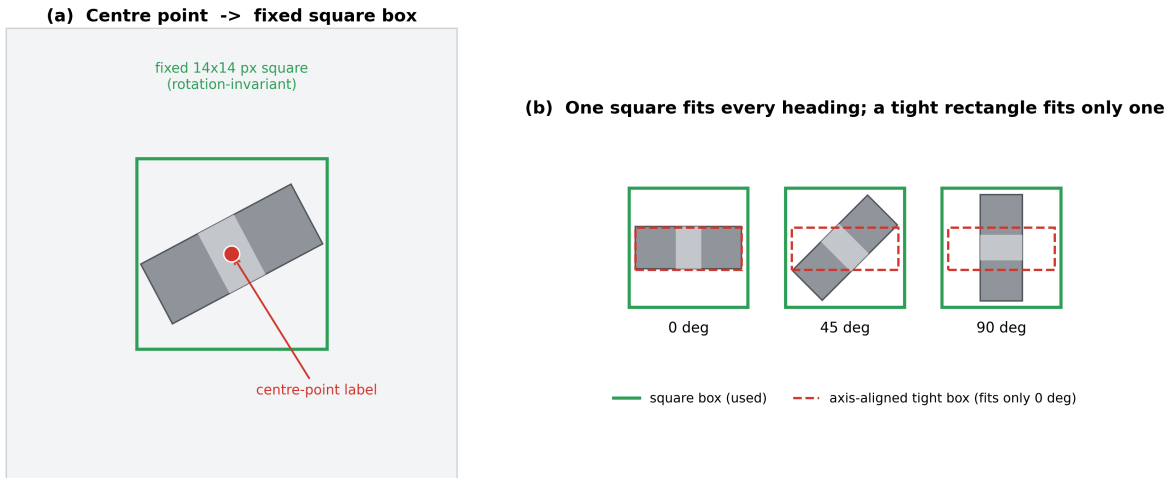


Figure 8: Conversion of COWC centroids to axis-aligned square bounding boxes. **(a)** Centering of a fixed 14×14 px box on each coordinate. **(b)** Rotation-invariant geometry accommodating arbitrary vehicle orientations. Sizing the box below the maximum vehicle length (18 px) prevents overlap in dense clusters, a trade-off that motivates future work on oriented bounding boxes.

3.3.2 Resolution matching

Each scene was downsampled to the 25 cm target: a linear scale of 0.6 for COWC (from 15 cm) and a comparable reduction for VEDAI (from 12.5 cm). Imagery was resampled using a bilinear filter, chosen to approximate the way a true 25 cm sensor integrates light over a larger ground area. Annotation coordinates were handled separately from the

imagery. For COWC, the centre points were extracted at native resolution and only the coordinates were then scaled, so that image and labels remained in lockstep. Critically, the annotation masks themselves were never resized, since interpolating the single-pixel vehicle dots against their background would have erased them. Resolution matching serves two purposes established earlier. Firstly, it maximises source–target relatedness and so reduces negative-transfer risk (Weiss et al. 2016), and secondly it places the training data within the resolution band shown to support high accuracy for YOLOv8 (Section 2.3).

3.3.3 Tiling

COWC’s scenes are far larger than a detector input, so each 25 cm scene was tiled into 640×640 images (Figure 9). A stride of 512 pixels was used, giving roughly 20% overlap between adjacent tiles, with an additional edge-flush tile on each axis so that the far edges of a scene were fully covered (Listing 2). Together, the overlap and edge-flush tiling guarantee that every vehicle appears in at least one tile, eliminating the boundary losses of naive non-overlapping tiling and following the slicing-aided inference strategy of Akyon et al. (2022). Near-uniform tiles arising from mosaic padding were skipped using a standard-deviation threshold, meanwhile, vehicle-free tiles that contained a labelled hard negative were retained with empty label files to suppress false positives, and lastly a bounded fraction of pure-background tiles was kept.

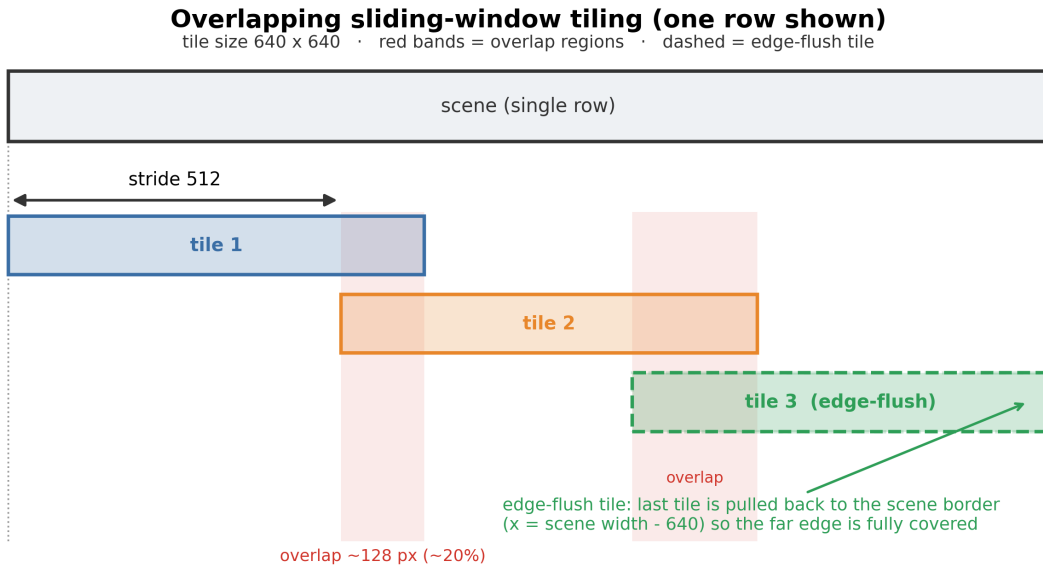


Figure 9: Overlapping sliding-window tiling scheme (single row shown). Tiles of 640×640 px are extracted at a 512 px stride to yield a 20% overlap (red). An edge-flush terminal tile ensures complete boundary coverage, guaranteeing all targets are captured within at least one tile under the Slicing-Aided Hyper Inference (SAHI) framework (Akyon et al. 2022).

3.3.4 Scene-level train/validation split

Training and validation data were separated at the scene level, never the tile level (Listing 3). This is because of the tiling overlap. A tile-level split would place near-

duplicate tiles together, which could potentially show the same vehicles in both sets, consequently inflating validation scores. One dense, RGB Toronto scene was held out for validation: RGB to match the colour domain of the UK imagery, dense to be representative of car-park scenes, and a whole, unseen scene to give an honest estimate of cross-scene generalisation. No COWC test split was created, because the genuine test set is reserved for UK retail imagery. This produced 2,948 tiles in total: 2,822 training tiles from 31 scenes and 126 validation tiles from the held-out scene, reconciling to the 37,721 unique annotated cars.

3.3.5 VEDAI-specific preparation

VEDAI was processed through the same logic, with three documented differences. First, only car-like classes (car, pickup, van, and the small- and large-vehicle meta-classes) were retained and collapsed to the single vehicle class, and all non-car classes (boats, planes, tractors, trucks and the residual “other” category) were dropped. Second, because normalised box coordinates are invariant under uniform downsampling, the boxes were computed at native pixel size and only the imagery was resampled to 25 cm, avoiding any coordinate drift. Third, no tiling was required, this is because each VEDAI frame is smaller than a single 640-pixel tile, so frames were letterboxed to 640 at training time, and an image-level random split (10% validation, fixed seed) was used because VEDAI frames are independent locations rather than one large mosaic. The result retained 1,581 car-like boxes and produced 1,121 training and 125 validation images, of which roughly 37% are legitimate negative (vehicle-free) frames that help suppress false positives.

3.4 Detector and training configuration

The detector was YOLOv8s, the small variant of the anchor-free, single-stage YOLOv8 architecture whose suitability for small-target aerial detection was argued in Section 2.5. It was fine-tuned from pretrained weights rather than trained from scratch: this is the transfer-learning route recommended for limited target data (Yosinski et al. 2014), and it matches the procedure validated by the resolution study this work builds on (Section 2.3).

The training configuration, held identical across the three source models so that results are comparable, is given in Table 2. Training was carried out on a UCL Computer Science GPU. The configuration: YOLOv8s, transfer learning, a 25 cm resolution and an input size of 640, deliberately mirrors Gliubičiūtė et al. (2023) so that the results are directly comparable with the resolution study that motivates the project’s core design decision.

Two single-source detectors were trained, one on COWC and one on VEDAI, under these identical settings, so that their within-domain accuracy and their behaviour on the other dataset could be compared directly.

Table 2: Training configuration, applied identically to the COWC, VEDAI and combined-source models.

Setting	Value
Model	YOLOv8s, pretrained weights (transfer learning)
Classes	1 (vehicle)
Input image size	640
Batch size	16
Epochs	up to 100, with early stopping (patience 20)
Optimiser	SGD with momentum (Sutskever et al. 2013) (selected automatically), initial learning rate 0.01
Hardware	NVIDIA RTX 4070 Ti SUPER (16 GB), CUDA

3.5 Combined-source model

To test whether pooling the two sources broadens coverage enough to reduce the domain gap (Weiss et al. 2016), a third detector was trained on the union of the two training tile sets (2,822 COWC and 1,121 VEDAI tiles), pointed at via a multi-dataset configuration so that no tiles were re-processed. Every other setting was held identical to the single-source runs, so that the training data was the only variable that changed. The combined model was then evaluated separately on each dataset’s validation set, allowing its per-domain performance to be compared against the corresponding specialist. This combined model is also the detector carried forward as the *transfer* model in the UK experiments (Section 3.9), where it is applied to the UK test sites with no further training.

3.6 Evaluation methodology

A single evaluation procedure was applied to all three source models. The reported metrics are precision (P), recall (R), and the F_1 score, together with mean Average Precision at an Intersection-over-Union (IoU) threshold of 0.5 (mAP@0.5) (Everingham et al. 2010) and over the stricter COCO range 0.5–0.95 (mAP@0.5 : 0.95) (Lin et al. 2014). These are the standard object-detection performance measures (Padilla et al. 2020), defined in Equations (1)–(4).

A predicted box counts as a true positive (TP) when its IoU with an unmatched ground-truth box exceeds the operating threshold, as a false positive (FP) otherwise, and an unmatched ground-truth box is a false negative (FN):

$$\text{IoU}(B_p, B_{\text{gt}}) = \frac{\text{area}(B_p \cap B_{\text{gt}})}{\text{area}(B_p \cup B_{\text{gt}})}, \quad (1)$$

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}, \quad (2)$$

$$AP = \int_0^1 P(R) dR, \quad mAP = \frac{1}{|C|} \sum_{c \in C} AP_c, \quad (3)$$

$$mAP@[0.5 : 0.95] = \frac{1}{10} \sum_{\tau \in \mathcal{T}} mAP@_{\tau}, \quad (4)$$

where $\mathcal{T} = \{0.50, 0.55, \dots, 0.95\}$.

With a single vehicle class ($|C| = 1$) the mean over classes is trivial, so mAP equals the per-class AP. mAP@0.5 uses the fixed IoU threshold of (1), while mAP@0.5 : 0.95 averages it over ten thresholds and so rewards tight localisation. The COCO-style breakdown of AP by object area (small, medium, large) is produced only when a COCO-format split is available; for the plain YOLO datasets used here it is not computed, and is recorded as unavailable in the results rather than estimated. In practice this is a minor limitation: at 25 cm a car spans only about 14–18 pixels, so essentially every vehicle falls below the 32×32 -threshold that the COCO evaluation protocol (Lin et al. 2014) uses to define a "small" object, so the overall mAP is already, in effect, an all-small score, which, as argued in Section 2.2, is precisely the quantity that matters for this task.

Each model was evaluated in two ways. The within-domain evaluation tested each model on its own held-out validation set, serving as a sanity check that the evaluator reproduces the training results. The cross-domain evaluation tested each single-source model on the other dataset’s validation set, providing a direct measure of generalisation and quantifying the marginal-distribution domain gap formalised in Section 2.6. All evaluation runs used identical thresholds (a confidence threshold of 0.001 and an IoU threshold of 0.6 for non-maximum suppression (Neubeck & Van Gool 2006)) so that the resulting figures are comparable. The operating confidence threshold for deployment was not fixed at the conventional 0.5 but selected at the peak of each model’s F1–confidence curve (Listing 5). The selected values are reported with the results, and the threshold matters for the counting step described next. Throughout, comparisons were restricted to runs sharing the same dataset, class set, resolution and IoU convention, in line with the comparability discipline emphasised in the object-detection evaluation literature (Everingham et al. 2010, Padilla et al. 2020).

3.7 UK retail dataset: site selection and annotation

The deployment domain is UK retail car-park imagery at 25 cm per pixel, obtained from the EDINA Aerial Digimap service (Getmapping) (Getmapping Plc 2021). A labelled UK dataset was constructed both to adapt the detector to the target domain and to provide the genuine held-out test toward which every earlier result was building.

3.7.1 Site selection

Thirty retail sites (UK001–UK030) were drawn from the GeoLytx Supermarket Retail Points open dataset (GEOLYTIX 2024) and recorded in a site registry. The complete registry is given in Table 9, in Appendix B. Each candidate was verified in Digimap to satisfy a consistent set of inclusion criteria: a surface car park visible in the aerial imagery, a definable car-park boundary, and an estimable parking capacity. For every accepted site the registry records the retailer and fascia, store name and location, coordinates, imagery resolution and *capture date*, an estimated capacity obtained by manually counting the visible marked bays, an occlusion level, and a car-park size class (Small, Medium or Large). Recording the capture date directly addresses the temporal-comparability weakness identified in open-satellite work (Section 2.8), since it allows each observation to be tied to a known acquisition. The selected sites span the major UK grocery retailers: Aldi, Asda, Iceland, Lidl, Marks & Spencer, Morrisons, Sainsbury’s, Tesco and Waitrose, so that the dataset is not dominated by the layout conventions of any single chain.

3.7.2 Per-site preparation

Each site was processed by the same standardised pipeline used for the public sources, adapted to the UK imagery. For a given site the raw Getmapping tiles were mosaicked and clipped to the car-park area of interest in QGIS (QGIS Development Team 2026), and cut into fixed 640×640 tiles at 25 cm per pixel, so that the UK tiles are dimensionally identical to the training tiles. A pretrained detector then pre-labelled each tile at a low confidence threshold to seed annotation. These machine proposals were used only as a starting point and carried no ground-truth status.

3.7.3 Annotation and ground truth

Every pre-labelled tile was manually corrected in the Computer Vision Annotation Tool (CVAT) (CVAT.ai Corporation 2026) to produce the ground-truth vehicle boxes (Figure 10). Corrected annotations were exported and converted to the project’s common YOLO form: each box (rotated boxes included) had its four corners projected, the axis-aligned bounding box taken, and its coordinates normalised against the tile under the single *vehicle* class, in the same YOLO label format as the training data.

Unlike the fixed-size square boxes synthesised for COWC (Figure 8), however, these UK boxes are tight axis-aligned boxes drawn to each vehicle: the label *format* is identical but the box *geometry* is not, and the consequence of this difference for the strict-IoU localisation metric is examined in the discussion.

For sites that required more than one tile, vehicles appearing in the overlap between adjacent tiles were de-duplicated using an intersection-over-union merge ($\text{IoU} = 0.5$), so that each physical vehicle is counted once at the site level. This produced a fully hand-verified UK ground-truth set of 30 annotated sites, comprising both single-tile and

multi-tile car parks.

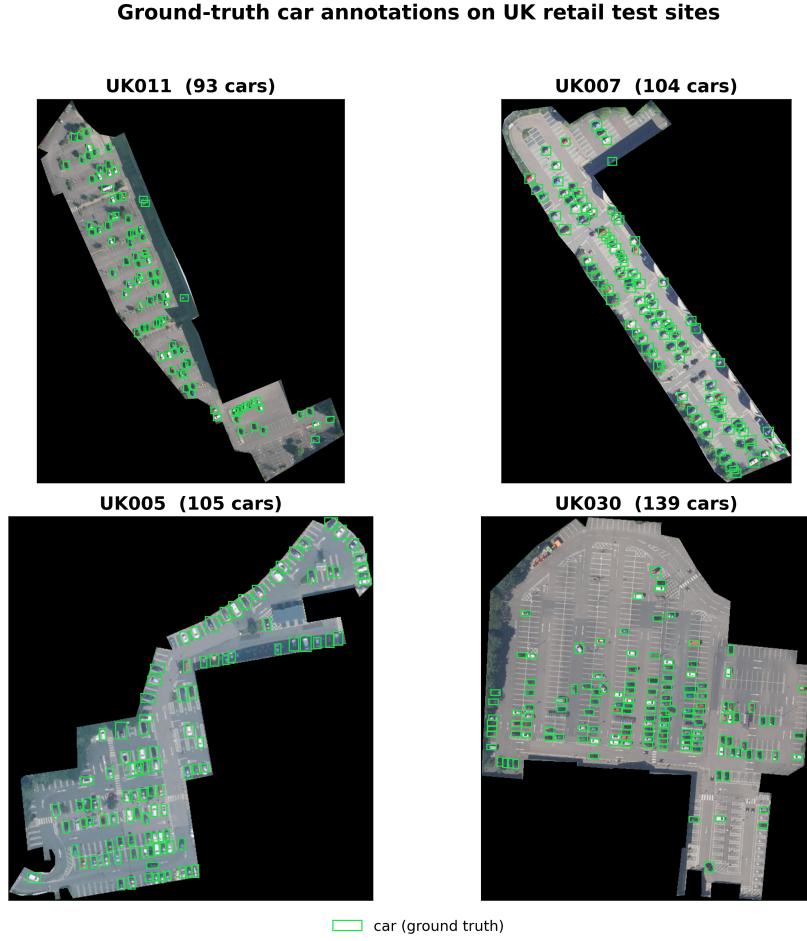


Figure 10: Hand-verified ground-truth vehicle annotations on four UK retail test tiles (25 cm per pixel, 640×640). Each green box is one manually corrected axis-aligned vehicle box; the count shown is per tile. The examples span sparse to densely occupied lots.

3.8 Dataset split for UK adaptation and evaluation

The thirty UK sites were partitioned into ten *calibration* sites, used for domain adaptation, and twenty *held-out test* sites, used only for evaluation. Two properties of this split are important for the validity of the results.

First, the split was made at the *site level*, never the tile level. As with the scene-level split of the public data (Section 3.3.4), this prevents overlapping tiles from a single car park from appearing in both the adaptation and the evaluation sets, which would otherwise inflate the reported accuracy. Every tile of a given site falls entirely on one side of the split.

Second, the split was *frozen before any detector was evaluated*, and in particular before any UK detection accuracy was known. It is a deliberate, balance-aware assignment: the calibration and test folds were each chosen to contain a spread of retailers and of small, medium and large car parks, so that neither fold is skewed toward a particular chain or

lot size (Figure 11).

The assignment is recorded directly in the `proposed_split` column of the site registry and read unchanged at every subsequent stage; no script randomises or re-derives it. The registry-finalisation step validates the frozen registry, confirming that every accepted site carries a split, a 25 cm resolution and a capture date. It re-derives each site’s size class from its estimated capacity (Small, fewer than 75 bays; Medium, 75–200; Large, more than 200), but it never alters the split, and the assembly step likewise only reads it. Since the assignment predates any UK detection result, the test set cannot have been chosen, consciously or otherwise, to favour sites on which the detector performs well.

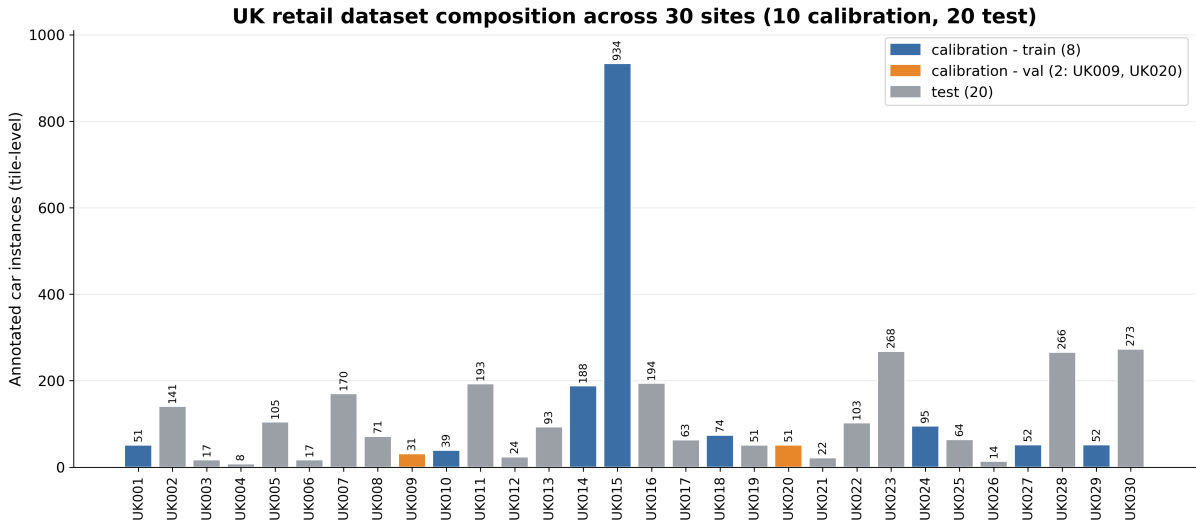


Figure 11: De-duplicated vehicle counts across the thirty UK study sites, categorized by fold: calibration–train (8 sites), calibration–val (UK009 and UK020), and held-out test (20 sites). The distribution confirms a balanced split across diverse lot sizes. Corresponding per-tile label counts are detailed in table 3.

The ten calibration sites are UK001, UK009, UK010, UK014, UK015, UK018, UK020, UK024, UK027 and UK029; the remaining twenty form the held-out test set. Within the calibration pool, two sites (one Small (UK009) and one Large (UK020)) were named as an internal validation set for monitoring the adaptation run, following the same explicit, named-split discipline used for the public data. The twenty test sites were never used for training or for monitoring at any point.

Assembling the UK data into the detector’s expected directory layout was scripted: the assembly step reads the frozen split from the registry, pairs each tile image with its converted label, excludes the annotation-overlay images and any tile with no vehicles, and writes the calibration and test image–label trees together with the internal train/validation tree used for adaptation. The resulting UK dataset contained the tile counts summarised in Table 3.

Table 3: Assembled UK retail dataset. Instance counts are per-tile label counts, which are the quantities used by the detector; they differ from the seam-de-duplicated site totals used for counting, because a vehicle in a tile overlap is represented in each tile it appears in.

Fold	Tiles	Vehicle boxes	Purpose
Calibration (10 sites)	15	1,567	UK adaptation pool
train (8 sites)	13	1,485	fine-tuning data
val (2 sites)	2	82	monitor fine-tuning (UK009, UK020)
Test (20 sites)	28	2,157	held-out evaluation

3.9 UK experiments: baseline, transfer, and domain adaptation

The UK evaluation was structured as three experiments of increasing adaptation, each evaluated on the *same* twenty held-out test sites, so that the contribution of each stage could be isolated against a single fixed benchmark (Figure 12).

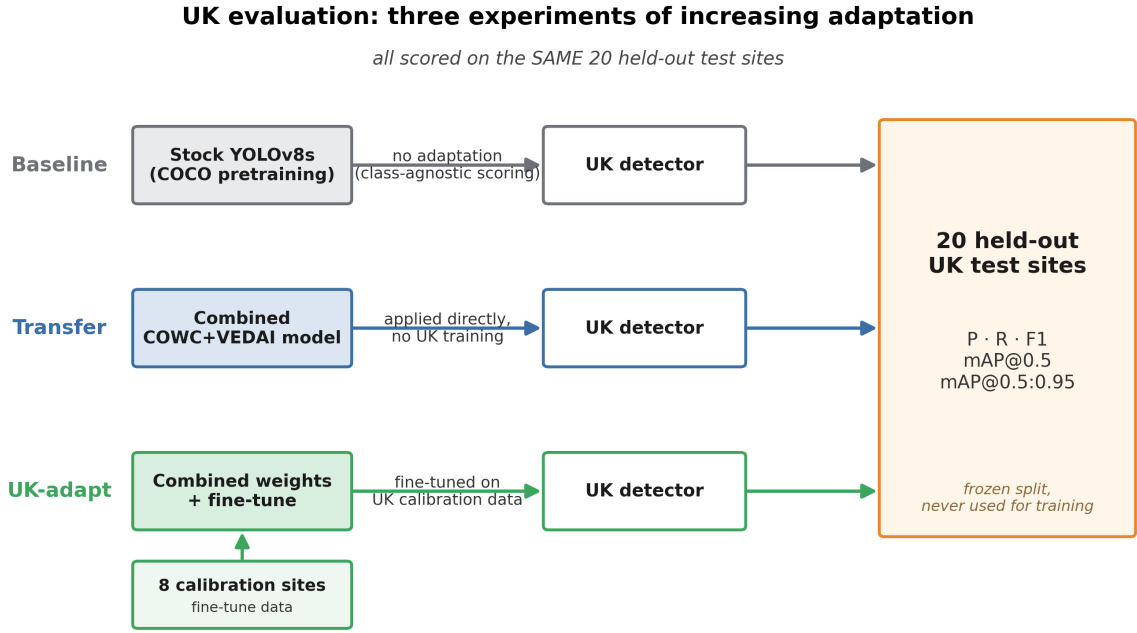


Figure 12: The three experimental configurations evaluated on the twenty held-out UK test sites. **Baseline:** Class-agnostic, unmodified YOLOv8s (COCO) detector. **Transfer:** Joint COWC+VEDAI source model applied zero-shot to UK target sites. **UK-adaptation:** Source model fine-tuned on the eight calibration–train sites. Evaluating all configurations against a single benchmark isolates the relative impacts of source pretraining versus local fine-tuning.

The three detectors were:

- **Baseline.** The stock YOLOv8s model with its original generic COCO (Lin et al. 2014) pretraining and no adaptation to overhead imagery. This establishes the performance

of an off-the-shelf detector on the task and provides the reference point against which adaptation is measured.

- **Transfer.** The combined COWC+VEDAI model of Section 3.5, applied directly to the UK test sites with no UK training. This measures pure cross-domain transfer from the resolution-matched public sources to UK retail imagery, and is the concrete realisation of the transfer-learning strategy set out in Section 2.6.
- **UK-adaptation.** The transfer model, further fine-tuned on the eight calibration–train sites and monitored on the two named validation sites. This measures the effect of adding a small quantity of labelled target data, and tests directly the prediction from transfer-learning theory that negative-transfer risk falls, and target accuracy rises, as in-domain labels are introduced (Section 2.6).

The stock model retains its original COCO output layer, which predicts eighty object classes rather than the single *vehicle* class used throughout this project; in the COCO label set, moreover, class 0 is *person*, with cars forming a separate class. Scored naively against the single-class UK ground truth, the baseline would be credited only for detections that happened to fall under the ground-truth class index and penalised for the genuine car detections it emits under COCO’s own *car* class. To remove this labelling artefact, and to give the off-the-shelf detector the most generous fair treatment. The baseline was evaluated *class-agnostically*, meaning that all predicted classes were collapsed to a single foreground category, so that any detection overlapping a true vehicle, counts as a match regardless of the COCO label attached to it (Listing 6). This measures exactly what the baseline is intended to establish, namely whether an unadapted detector can localise cars at all. The adjustment is a no-op for the transfer and UK-adaptation models, which are already single-class *vehicle* detectors, so all three experiments are scored on an identical and directly comparable footing.

The adaptation run reused the project’s standard training configuration (Table 2), fine-tuning from the combined-source weights at the native tile size of 640 pixels, with early stopping on the validation set. As the UK adaptation was carried out on a different machine from the source-model training, it ran on an NVIDIA RTX 3090 Ti (24 GB) rather than the card used for the public-data training; the training procedure was otherwise unchanged, with the optimiser again selected automatically.¹

All three detectors were scored with the single evaluation procedure of Section 3.6, reporting precision, recall, F1, mAP@0.5 and mAP@0.5:0.95 on the twenty held-out test

¹The automatic optimiser selection chose AdamW (Loshchilov & Hutter 2019) (initial learning rate ≈ 0.002) for the UK fine-tune, whereas the source-model training selected SGD; this is the one respect in which the otherwise identical configuration differed, and it follows from the smaller adaptation set.

sites under identical thresholds. The results of these three experiments, which constitute the project’s principal detection findings, are reported in the following chapter.

3.10 Occupancy estimation

With the detector adapted to and validated on UK imagery, the final stage of the method converted its detections into an estimate of car-park occupancy. This stage built directly on the adapted detector established above.

Detections across a site’s tiles were merged into a single site-level count using the same seam-aware intersection-over-union procedure already used to de-duplicate the ground truth (Listing 4). This mirrors the standard practice in sliced or tiled inference, where the detections produced on overlapping image slices are reconciled by an overlap criterion before they are counted (Akyon et al. 2022). Occupancy was then estimated as the ratio of the site-level vehicle count (obtained by detection-based counting, as in drone and overhead car-park counting benchmarks (Hsieh et al. 2017, Mundhenk et al. 2016)) to the site’s total parking capacity, taken to be each site’s estimated capacity as recorded in the site registry (Section 3.7). Detections were thresholded at the model’s F1-optimal operating point rather than the conventional 0.5; that choice is deliberate, because the confidence threshold that maximises F1 is a property of the particular model and data rather than a fixed constant and so must be selected empirically (Lipton et al. 2014). The threshold was fixed on the held-out calibration-monitoring sites and applied unchanged to the test sites, so it was never tuned on the data it was evaluated against.

Counting quality was reported *separately* from detection quality, in deliberate contrast to the prior UK work that conflated the two (Section 2.8). For a site i with predicted count $\hat{c}_i$, true count c_i and capacity K_i , occupancy is $o_i = c_i/K_i$ (predicted $\hat{o}_i = \hat{c}_i/K_i$), and over the N test sites the count error is summarised by the mean absolute error, root-mean-square error, bias, mean absolute percentage error and coefficient of determination:

$$\text{MAE} = \frac{1}{N} \sum_i |\hat{c}_i - c_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_i (\hat{c}_i - c_i)^2}, \quad \text{Bias} = \frac{1}{N} \sum_i (\hat{c}_i - c_i), \quad (5)$$

$$\text{MAPE} = \frac{100}{N} \sum_i \frac{|\hat{c}_i - c_i|}{c_i}, \quad R^2 = 1 - \frac{\sum_i (\hat{c}_i - c_i)^2}{\sum_i (c_i - \bar{c})^2}, \quad (6)$$

where $\bar{c}$ is the mean true count.

The coefficient of determination is taken against the identity ($y = x$) line, so it penalises systematic over- or under-counting rather than merely correlation. The same measures are reported for the occupancy rate o_i . Reporting these counting metrics apart from the detector’s P , R , F_1 and mAP is the methodological correction that distinguishes this project from its closest precedent. It ensures that a good count arising from compensating detection errors cannot be mistaken for accurate detection.

3.11 Spatial aggregation of detections

Each detection carries a location, therefore the site-level counts can be re-expressed as spatial data. For the analysis presented in the discussion (Section 5.4), every detected box on a test site was reduced to its centroid and projected from tile-local pixels into British National Grid coordinates (EPSG:27700), using the per-tile geo-referencing recorded during preparation (each tile’s ground origin and pixel size; see Listing 7). The resulting per-site point set was exported in a GIS-ready vector format for overlay with contextual layers, and a continuous parking-density surface was formed by Gaussian kernel density estimation over the centroids with a 6 m bandwidth (Silverman 1986). This yields, for each site, a georeferenced map of where parking demand concentrates. Unlike the detection and counting outputs, the density surface is not scored against ground truth. It is therefore used qualitatively, and its interpretation and limitations are set out in Section 5.4.

3.12 Entrance-proximity analysis of parking demand

The density surface of Section 3.11 shows *where* demand concentrates within a lot, but not relative to any feature of the site. For retail parking a more specific question is whether that concentration is organised around the points of access, namely the store’s pedestrian entrance and the vehicle entrance to the car park. This can be tested directly by relating the position of each detected vehicle to the position of the entrances, converting the qualitative observation that “occupancy has a location” (Section 5.4) into a measured, per-site statistic because every detection is georeferenced, .

For each of the twenty held-out test sites, two entrance types were digitised by hand in QGIS (QGIS Development Team 2026) over the georeferenced site clip: the principal retail (pedestrian) entrance of the store, placed at the visible storefront, and the principal vehicle entrance to the lot, placed at the junction between the access road and the car park. Each was recorded as a point layer in the same EPSG:27700 GeoPackage that holds the detected vehicle centroids and the car-park boundary, or area of interest (AOI). Manual digitisation over the imagery was used in preference to an external source because open map data does not reliably record these features for UK retail sites. The georeferencing established in Section 3.11 ensures the digitised points share a common coordinate frame with the detections.

For a site i and entrance type $t \in \{\text{retail}, \text{carpark}\}$, let E_i^t be the set of digitised entrance points, $\{c_{ij}\}$ the detected vehicle centroids and $\{g_{ik}\}$ a set of reference points sampling the available parking space. The proximity of any point x to the entrance is its nearest Euclidean distance,

$$d(x; E_i^t) = \min_{e \in E_i^t} \|x - e\|_2. \quad (7)$$

A raw summary of the vehicle distances $d(c_{ij}; E_i^t)$ is not directly interpretable, because it is confounded by the geometry of the lot: an elongated car park places most of its parking

far from any single entrance, so vehicles would appear “far” even under a strong preference to park close. To separate driver preference from lot shape, the vehicle distances are compared against the distances that the *available parking space* itself has to the entrance. The AOI polygon was sampled on a regular 2 m grid to yield the reference points $\{g_{ik}\}$ spanning the parkable area, and the same nearest-distance (7) was evaluated for each. The grid spacing, well below a single bay width, is fine enough that the reference distribution is insensitive to it. It plays the role of an inhomogeneous null, describing where vehicles would fall relative to the entrance if parking were spread uniformly across the lot.

Two statistics summarise the comparison. The first is the *distance ratio*, the mean vehicle distance expressed as a fraction of the mean available-space distance,

$$r_{\text{dist}} = \frac{\frac{1}{N_c} \sum_j d(c_{ij}; E_i^t)}{\frac{1}{N_g} \sum_k d(g_{ik}; E_i^t)}, \quad (8)$$

where N_c and N_g are the numbers of vehicles and reference points. A value $r_{\text{dist}} < 1$ means vehicles sit nearer the entrance than the average parking space does (demand clusters toward the entrance); $r_{\text{dist}} \approx 1$ indicates no preference, and $r_{\text{dist}} > 1$ a tendency to park away from it. The second statistic resolves this into a gradient. The distance range is partitioned into concentric rings of width $w = 10$ m, and for ring b the *relative fill* is the share of vehicles it contains divided by the share of available space it contains,

$$\text{RF}_b = \frac{n_b^c/N_c}{n_b^g/N_g}, \quad (9)$$

with n_b^c and n_b^g the counts of vehicles and reference points in the ring. $\text{RF}_b > 1$ marks a ring that is fuller than the lot average; a value that declines monotonically with distance is the signature of a lot that fills from the entrance outward. The strength of that decline is summarised by the Spearman rank correlation ρ_s between the ring mid-distance $\bar{d}_b$ and RF_b , a negative value indicating that fill falls as distance from the entrance increases.

Both statistics were computed per site and per entrance type from the adapted detector’s centroids (Listing 8), and are reported across the twenty test sites in Section 5.4. Three properties of the measure should be kept in mind and are revisited in the discussion. The denominator in (8)–(9) is the available *area* of the lot rather than its marked bays, a proxy that assumes parking is possible across the AOI. Distances are straight-line rather than walking routes, and the entrance points are a single hand-placed judgement per type, based on the storefront rather than an independently surveyed location. As with the density surface, the analysis rests on a single-capture snapshot, so it characterises the spatial structure of one observation rather than its evolution over time.

3.13 Experimental setup and reproducibility

This subsection consolidates the information needed to reproduce the runs reported in this dissertation. The detector and its training hyperparameters are given in Table 2, and the UK-adaptation settings in Section 3.9; what follows records only the software environment, the random seeds and determinism settings, the exact model checkpoint, and the availability of code and data. A complete pinned dependency list is provided in Appendix 6.3.

Software environment. Model training and inference ran on the two NVIDIA GPU machines of Table 2 and Section 3.9; data preparation, annotation and spatial analysis ran locally on macOS. The detector is the Ultralytics implementation of YOLOv8 (Jocher et al. 2023), pinned to release 8.2.0 for every run, so that training and inference behaviour is stable across re-runs. Both environments are listed in Table 4.

Model checkpoint. Every source model was initialised from the public `yolo8s.pt` checkpoint (COCO-pretrained, three-channel RGB input), and the UK-adaptation model from the combined-source weights, consistent with the transfer-learning route of Section 3.4. The remaining hyperparameters are exactly those of Table 2; automatic mixed-precision training was enabled (`amp = true`), and data-augmentation settings were left at the Ultralytics 8.2.0 defaults, with mosaic augmentation disabled for the final ten epochs (`close_mosaic = 10`).

Random seeds and determinism. A single fixed seed (`seed = 0`) and deterministic mode (`deterministic = true`) were set for every run, governing weight initialisation, data-loader shuffling and the image-level VEDAI split (Section 3.2). The scene-level and site-level splits are not random and carry no seed, since they are frozen assignments read from configuration and from the site registry (Sections 3.3.4 and 3.8). Even with these settings, exact bitwise reproduction is not guaranteed, because some GPU operations remain non-deterministic and the source models and the UK fine-tune were trained on different cards (Section 3.9). Each configuration was trained once rather than averaged over multiple seeds, so the reported figures are single-run point estimates; repeating the runs under different seeds to quantify run-to-run variance is noted as future work (Section 6.3).

Code and data availability. The full pipeline, comprising the data-preparation components (Section 3.3), the training and evaluation configurations, the per-run artefacts (`args.yaml` and `results.csv` for each model), the occupancy-estimation and entrance-proximity analysis code (Sections 3.10 and 3.12), and the site registry (Appendix B), is available at <https://github.com/paulaxho/carpark-project>, released under the MIT

licence. The two public training sources are obtainable from their original providers: COWC (Mundhenk et al. 2016) and VEDAI (Razakarivony & Jurie 2016). The UK site list derives from the open GeoLytix Supermarket Retail Points dataset (GEOLYTIX 2024). The UK aerial imagery, however, is licensed through UCL via the EDINA Aerial Digimap (Getmapping) service (Getmapping Plc 2021) and *cannot be redistributed*; the repository therefore provides the code, the hand-verified UK vehicle annotations in YOLO format, and the digitised entrance and car-park-boundary layers, together with the exact tile identifiers and capture dates needed to re-download the corresponding imagery through Digimap. With imagery so obtained, every result in this dissertation can be regenerated from the released code.

Table 4: Software environment. Model training and inference ran on the GPU machines of Table 2 and Section 3.9; data preparation, annotation and spatial analysis ran locally. The complete pinned export is in Appendix 6.3.

Component	Version
<i>Model training and inference (NVIDIA GPU, Linux)</i>	
Operating system	Rocky Linux 9.8 (Blue Onyx)
Python	3.9.25
PyTorch	2.3.0+cu121
torchvision	0.18.0
CUDA build	12.1
cuDNN	8.9.2.26
NVIDIA driver	<RTX 4070 Ti SUPER box>; <RTX 3090 Ti box>
Ultralytics (YOLOv8)	8.2.0
<i>Data preparation and spatial analysis (macOS)</i>	
Operating system	macOS 26.3.1
Python	3.11.15
NumPy	1.26.4
OpenCV	4.9.0.80
rasterio	1.3.10
GDAL	3.13.0
Shapely	2.0.4
GeoPandas	0.14.4
SciPy	1.17.1

4 Results

This chapter reports the project’s empirical findings. All results follow the evaluation procedure of Section 3.6. The source models are evaluated within and across the two public domains (Section 4.1); the three UK detectors of increasing adaptation are then evaluated on the twenty held-out UK test sites (Section 4.2), which constitute the project’s principal detection findings; and finally those detectors are turned into car counts and occupancy estimates on the same test sites (Section 4.4), the applied outcome the project set out to test.

4.1 Source models: within- and cross-domain performance

Each single-source detector performs strongly on its own validation set. The COWC model reaches an mAP@0.5 of 0.934 (recall 0.886) and the VEDAI model 0.818 (recall 0.714); these within-domain scores, summarised in Table 5, confirm that the training pipeline reproduces the accuracies reported by the resolution study it mirrors (Section 2.3), and they serve as the reference against which generalisation is measured.

Table 5: Source-model results: each model evaluated within and across the two public domains. Precision, recall and F1 are taken at each model’s peak-F1 operating point.

Trained on	Evaluated on	P	R	F1	mAP@0.5	mAP@0.5:0.95
COWC	COWC (val)	0.924	0.886	0.905	0.934	0.487
COWC	VEDAI (val)	0.311	0.424	0.359	0.202	0.045
VEDAI	VEDAI (val)	0.848	0.714	0.775	0.818	0.479
VEDAI	COWC (val)	0.230	0.258	0.243	0.168	0.036
Combined	COWC (val)	0.938	0.894	0.916	0.943	0.516
Combined	VEDAI (val)	0.640	0.692	0.665	0.696	0.344

Applied to the other domain, both specialists collapse. The COWC model scores only 0.202 mAP@0.5 on VEDAI, and the VEDAI model only 0.168 on COWC, roughly a fifth of their within-domain accuracy, and even lower under the strict mAP@0.5:0.95 (0.045 and 0.036). This is the marginal-distribution domain gap formalised in Section 2.6 made concrete. A detector trained on one overhead-vehicle distribution does not, on its own, transfer to another. This is a well documented pattern in aerial vehicle detection. Fang et al. (2025) report AP_{50} losses of comparable magnitude when detectors trained on one geographic or sensor domain are evaluated on another, and motivate their adaptation method precisely on this failure to generalise across overhead-vehicle domains. The gap is also asymmetric: VEDAI→COWC (0.168) is worse than COWC→VEDAI (0.202), consistent with the argument of Section 2.4 that VEDAI’s deliberately sparse scenes are poor preparation for COWC’s parking lots. Whereas the denser source transfers slightly

better in the reverse direction.

Pooling the two sources recovers cross-domain coverage. The combined model matches the COWC specialist on COWC (0.943 versus 0.934 mAP@0.5) and, crucially, lifts VEDAI-domain accuracy from the 0.202 of the COWC specialist to 0.696. A gain of 0.49, at only a modest cost to within-domain VEDAI performance relative to the VEDAI specialist (0.696 versus 0.818). Figure 13 shows this pattern at a glance: strong on the diagonal, collapsed off it for the specialists, and high on both columns for the combined model. This is due to the retention of near-specialist COWC accuracy while regaining most of the VEDAI domain. The combined model is the one carried forward as the *transfer* detector in the UK experiments (Section 3.9).

Cross-domain generalisation of the source models

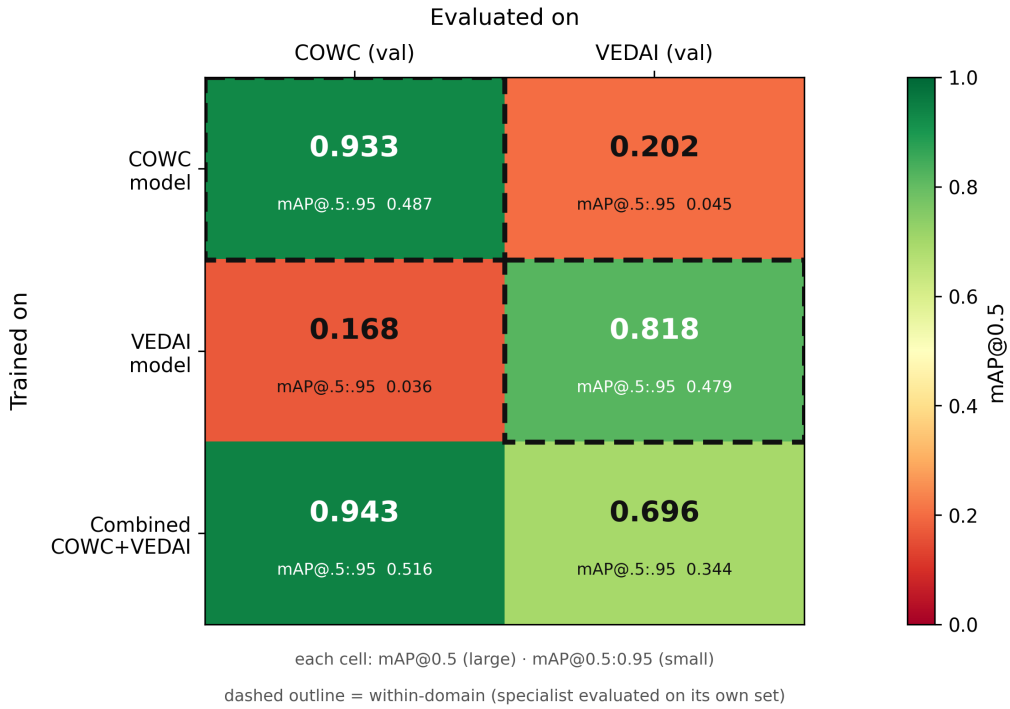


Figure 13: Evaluation matrix of source models (rows) across validation sets (columns) measuring mAP@0.5 and mAP@0.5:0.95. Outlined cells denote within-domain baselines. Single-source models exhibit severe off-diagonal performance degradation, whereas the combined model maintains strong generalisation across both domains.

4.2 UK experiments: baseline, transfer and domain adaptation

The three UK detectors were scored on the same twenty held-out test sites; Table 6 and Figure 14 report the results. The stock YOLOv8s baseline, scored class-agnostically so that it is credited for any detection on a real vehicle regardless of its COCO label (Section 3.9), reaches an mAP@0.5 of only 0.267. It localises just under half the vehicles (recall 0.466) but at a precision of 0.257, so most of its detections are false. This is the

floor that an off-the-shelf detector achieves on 25 cm UK retail imagery.

Table 6: Detection performance across the twenty held-out UK test sites. Precision (P), recall (R), and F_1 are evaluated at each model’s peak- F_1 operating threshold.

Experiment	P	R	F1	mAP@0.5	mAP@0.5:0.95
Baseline (stock YOLOv8s)	0.257	0.466	0.331	0.267	0.091
Transfer (COWC+VEDAI)	0.695	0.483	0.570	0.528	0.146
UK-adaptation (fine-tuned)	0.846	0.845	0.845	0.886	0.352

Transferring the combined COWC+VEDAI model to the UK, with no UK training, roughly doubles this to 0.528 mAP@0.5. The gain is almost entirely in precision, which rises from 0.257 to 0.695 at essentially unchanged recall (0.483). Overhead-vehicle pretraining teaches the detector to reject the false positives that dominate the baseline, but it does not, on its own, find substantially more UK cars. Indeed the transfer model’s recall is capped near 0.59 even at zero confidence. More than 40% of UK vehicles are simply never proposed by a model trained only on non-UK data, reflecting the appearance and density gap between the public sources and UK retail lots.

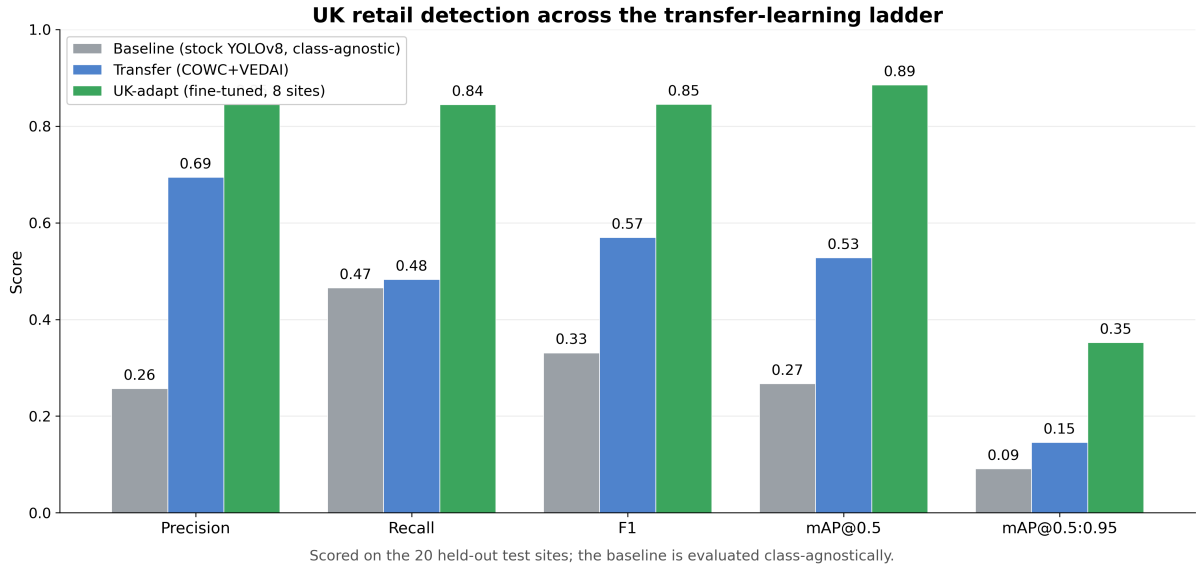


Figure 14: Detection metrics across the twenty held-out UK test sites. Incremental adaptation increases mAP@0.5 from 0.267 (baseline) to 0.528 (zero-shot transfer) and 0.886 (fine-tuned). The transfer phase primarily improves precision under near-constant recall, whereas subsequent fine-tuning enhances both metrics.

Fine-tuning that model on the eight UK calibration sites closes most of the remaining gap. UK-adaptation reaches an mAP@0.5 of 0.886, with precision and recall now balanced at 0.846 and 0.845. Recall nearly doubles relative to transfer (0.483 to 0.845): the vehicles that were invisible to the non-UK model become detectable once a small amount of

in-domain data is seen. Its recall ceiling rises from 0.59 to 0.94, while precision is retained. The most striking feature of Figure 14 is the size of this final step. Fine-tuning on just eight sites adds 0.36 to $\text{mAP}@0.5$, more than the 0.26 that the entire source-pretraining stage added over the baseline. For this target domain, a small, carefully held-out calibration set is worth more than a large volume of related but out-of-domain data.

Figure 15 shows what these numbers look like on a single representative site. The stock baseline not only misses and duplicates vehicles but imposes its generic-object priors on the scene, drawing large boxes that enclose whole rows or blocks of the car park as single “vehicles”. This is a scale error that a detector trained on close-range COCO objects makes when confronted with tiny overhead cars. The transfer model has learned the correct object scale and boxes individual cars, but at its low operating point still floods the lot with false positives while missing genuine vehicles. Only the adapted model produces one tight box per car, with correct detections dominating and few false alarms or misses.



Figure 15: Detection on representative site UK023. Bounding boxes denote correct detections (green), false alarms (red), and missed vehicles (amber); banners report site counts and $P/R/F_1$. (a) Baseline severely over-predicts (313 vs. 146 true) with oversized boxes. (b) Transfer targets individual vehicles but yields high false positives. (c) UK-adaptation produces precise, one-to-one detections. See Table 6 for aggregate metrics across all test sites.

Two features of Table 6 are developed in the sections that follow. First, the ordering under the strict $\text{mAP}@0.5:0.95$ (0.091, 0.146, 0.352) follows the same ladder but at much lower absolute values, indicating that localisation remains loose even for the adapted model; this is examined in Section 5.1. Second, the adapted model’s UK accuracy (0.886) approaches the source models’ within-domain scores, which places the domain gap and its recovery in a single view (Section 4.3).

4.3 Domain gap and recovery

Placing the source and UK results side by side reveals a single, coherent domain-gap story. Figure 16 tracks one detector, the combined COWC+VEDAI model, as the evaluation domain moves away from its training distribution. On COWC, its own densest source, it scores 0.943 $\text{mAP}@0.5$; on VEDAI it scored 0.696; and lastly on the UK retail sites, with no UK training, 0.528. This descending gradient is the marginal-distribution shift of Section 2.6 measured in accuracy. COWC is effectively in-domain, VEDAI a related sensor,

and UK retail the most out-of-distribution of the three, with a drop of 0.41 mAP@0.5 from the in-domain COWC to zero-shot UK.

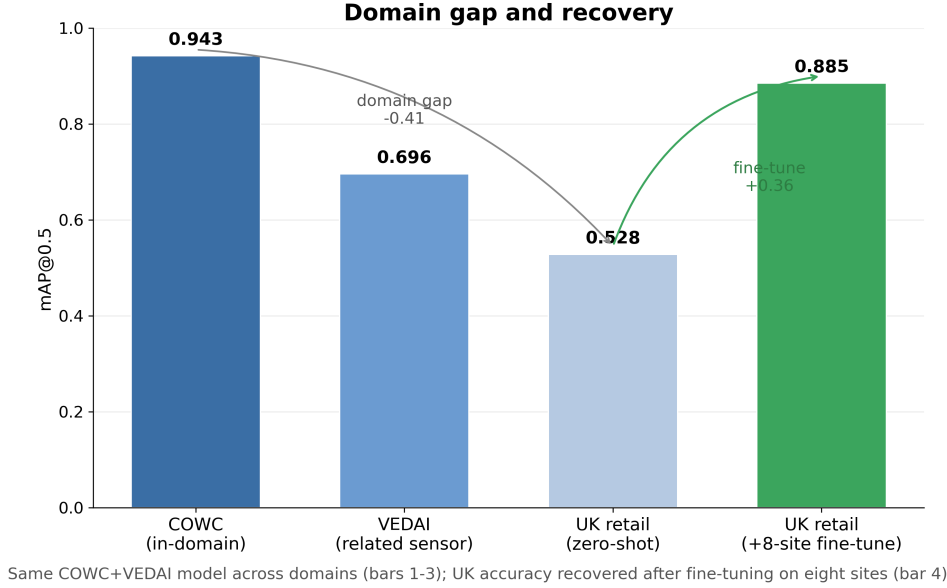


Figure 16: The combined COWC+VEDAI model’s mAP@0.5 as the evaluation domain moves away from its training distribution: COWC (in-domain, 0.943), VEDAI (related sensor, 0.696) and UK retail (zero-shot, 0.528) — a drop of 0.41. Fine-tuning the same model on eight UK calibration sites recovers UK accuracy to 0.886 (right-hand bar), approaching the in-domain score.

Fine-tuning reverses most of this drop. Adapting the same model on only eight UK calibration sites lifts UK accuracy from 0.528 to 0.886. A recovery of 0.36 of the 0.41 gap, so that the adapted UK detector not only exceeds the model’s own VEDAI-domain score (0.696) but approaches its in-domain COWC score (0.943). In effect, eight labelled sites convert an out-of-domain detector into a near-in-domain one. The magnitude of this recovery from a small target set is consistent with the transfer-learning literature (Weiss et al. 2016), in which the marginal value of additional source data diminishes once the target distribution diverges and a modest number of in-domain labels restores most of the lost accuracy. Fang et al. (2025) observe the same asymmetry between abundant source and scarce target data in the aerial-vehicle setting. As a small quantity of in-domain labels is introduced, target accuracy rises, and the risk of negative transfer recedes.

Two conclusions follow. First, the domain gap between public overhead-vehicle data and UK retail car parks is real and large. Essentially, this means that without adaptation, a model that is near-perfect on its source loses two-fifths of its accuracy on the target. Second, that gap is inexpensive to close, a calibration set of eight sites, held out at the site level and frozen before evaluation, recovers most of it. For a target domain with no public labelled data, this is the central practical finding of the project: a modest, carefully constructed in-domain set is both necessary and, on this evidence, largely sufficient to reach near-in-domain performance.

4.4 Counting and occupancy estimation

The detection results measure how well each model finds and localises vehicles, however the operational question is how well it *counts* them, and hence how well it estimates a site’s occupancy. Each model was run over the twenty held-out test sites at a fixed operating confidence, its per-tile boxes were de-duplicated across tile seams (Section 3.7) to give a single count per site, and occupancy was taken as that count divided by the site’s estimated capacity. Counting error is reported against the hand-annotated truth, which is directly verifiable, and occupancy-rate error alongside it. The operating threshold for each model was fixed in advance as the confidence that maximises F1 on the two calibration-monitoring sites, never on the test sites.

That threshold is itself diagnostic: it sits at a decisive 0.48 for the adapted model (F1 0.90) but collapses to near zero for the baseline and transfer models (0.004 and 0.012, F1 0.32 and 0.41), which have no clean operating point on UK imagery.

Table 7: Counting and occupancy performance on the twenty held-out UK test sites. Bias is predicted – true; R^2 is computed relative to the $y = x$ line. All models evaluated at calibration-optimized F_1 thresholds.

Model	Count (cars)				Occupancy	
	MAE	RMSE	Bias	R^2	MAE (pp)	R^2
Baseline (stock YOLOv8s)	66.2	96.6	+31.6	−3.40	27.9	−1.71
Transfer (COWC+VEDAI)	74.8	95.5	+69.2	−3.29	34.2	−2.44
UK-adaptation (fine-tuned)	5.4	8.1	+1.4	0.97	2.6	0.97

The adapted model counts accurately. Its mean absolute error is 5.4 cars per site against true counts spanning 8 to 146 vehicles, and its predicted counts track the truth along the identity line with $R^2 = 0.97$ (Figure 17). The count is within five cars on fifteen of the twenty sites and within ten on seventeen. The full per-site counts for all three models are listed in Table 10 (Appendix C). This carries directly into occupancy, which is recovered to within 2.6 percentage points of capacity on average, again with $R^2 = 0.97$ (Figure 18). The small positive bias (+1.4 cars) reflects a mild tendency to over-count, seen mainly on the larger lots. The two biggest residuals: UK028 (162 predicted against 138 true, the largest-capacity site) and UK013 (79 against 93, a densely packed and occluded lot at 93% occupancy), fall at opposite ends, over- and under-counting respectively under the conditions where tight localisation is hardest (Section 5.1).

The baseline and transfer models, by contrast, cannot count. Both post negative R^2 on counts and on occupancy, meaning each is a worse estimator than simply predicting the average site count. The failure is not subtle: their occupancy estimates miss by 28 and 34

percentage points on average, and both scatter far off the identity line in either direction (Figure 17). The transfer model is the more revealing case. It is a genuine detector, it scored 0.528 mAP@0.5 on these sites (Table 6), yet its counts carry a systematic +69-car bias, because at its low operating threshold it emits many spurious boxes while still missing the vehicles its capped recall never proposes, and these two errors do not cancel. A respectable detection score therefore does not translate into a usable count.

This is a recognised hazard of counting-by-detection rather than a quirk of these models. Lempitsky & Zisserman (2010) motivate density-based counting on exactly this ground, that per-object false positives and false negatives accumulate into the total instead of cancelling, and Di Mauro et al. (2019) likewise treat parking-occupancy estimation as a counting task evaluated in its own right rather than inferred from detection quality.

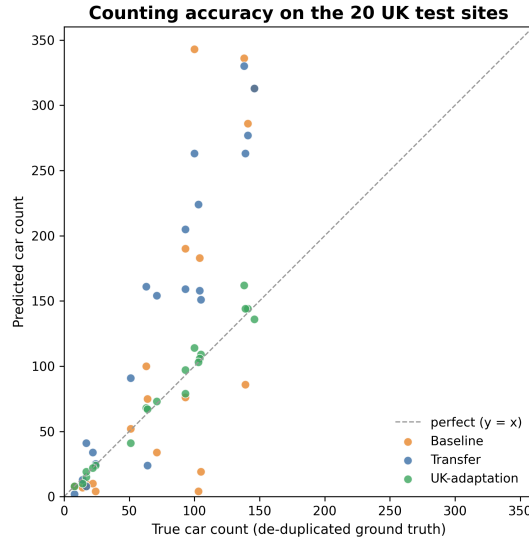


Figure 17: Predicted against true car count for the three models on the twenty held-out test sites.

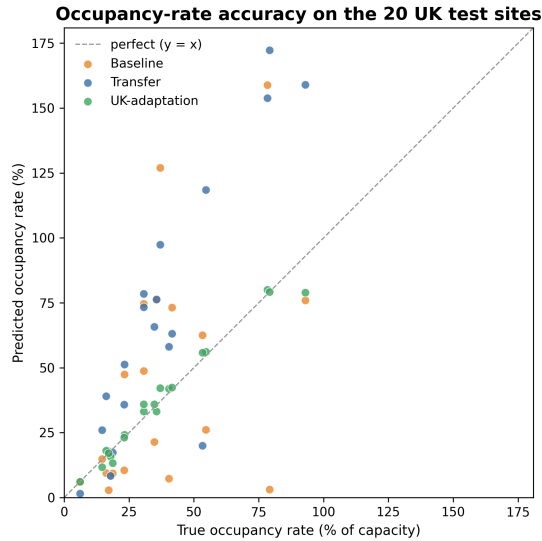


Figure 18: Predicted against true occupancy rate (car count as a percentage of estimated capacity).

Two qualifications frame these figures. First, the occupancy denominator is a capacity obtained by manually counting each site’s marked bays from the imagery: a direct count rather than a visual estimate, but not error-free where markings are faded or occluded, so the occupancy rate carries a denominator uncertainty that the verifiable vehicle count does not. This is why the counting metric is reported alongside it rather than occupancy alone. Second, occupancy is defined here as detected cars over capacity and can exceed 100% on lots where vehicles park in aisles or loading areas, so it measures utilisation rather than a strict fraction of marked bays. With those considerations in mind, the adapted detector supports occupancy estimation at single-site accuracy of a few percentage points, which is the applied outcome the project set out to test.

4.5 Entrance-proximity of parking demand

While counting and occupancy estimations reduce each site to a single scalar total, the localized detection centroids allow for a more granular spatial analysis: where within a lot does demand concentrate relative to points of access?

To measure this, the entrance-proximity framework (Section 3.12) was applied to the adapted detector’s centroids across the twenty test sites. Each site was evaluated using two statistics per entrance type:

1. **Distance ratio** (r_{dist}): Calculated via Equation (8), a ratio below 1.0 indicates that vehicles park closer to the entrance than the average available space does.
2. **Spearman gradient** (ρ_s): Measures the correlation between distance and relative fill; a negative value indicates that the lot fills progressively from the entrance outward.

The aggregate results across all sites are summarized in Table 8, with complete per-site values listed in Table 11 (Appendix C).

Table 8: Entrance-proximity of parking demand on the twenty held-out test sites, by entrance type. $r_{\text{dist}} < 1$ indicates vehicles closer to the entrance than the average available parking space; $\rho_s < 0$ indicates relative fill that declines with distance from the entrance.

Entrance type	Sites with $r_{\text{dist}} < 1$	mean r_{dist}	mean ρ_s
Retail (pedestrian)	18/20	0.88	−0.30
Car park (vehicle)	12/20	0.97	+0.02

The two entrance types behave very differently. Relative to the *retail* entrance, parking is consistently drawn toward the store: eighteen of the twenty sites record $r_{\text{dist}} < 1$, with a mean ratio of 0.88, and the relative fill declines with distance ($\rho_s < 0$) on the same eighteen, mean −0.30. Figure 19 shows this as an ordered sweep across sites, the great majority sitting left of the parity line. The effect is strongest at UK019 ($r_{\text{dist}} = 0.71$,

$\rho_s = -0.53$), UK003 (0.75) and UK030 (0.75, $\rho_s = -0.60$), where vehicles pack markedly toward the storefront while bays further out remain comparatively empty. The per-site gradients (Figure 20) make the mechanism explicit: relative fill is highest in the rings nearest the entrance and falls away with distance, most cleanly at sites such as UK013 ($\rho_s = -0.92$).

Only two sites run counter to this pattern, both with $r_{\text{dist}} > 1$: UK012 (1.12) and UK004 (1.24). UK004 carries just eight detected vehicles, so its ratio rests on very few points and should be read with the same caution applied to percentage errors on the smallest lots (Section 4.4); UK012 is a genuine exception in which demand sits away from the digitised entrance. Both are examined in Section 5.4.

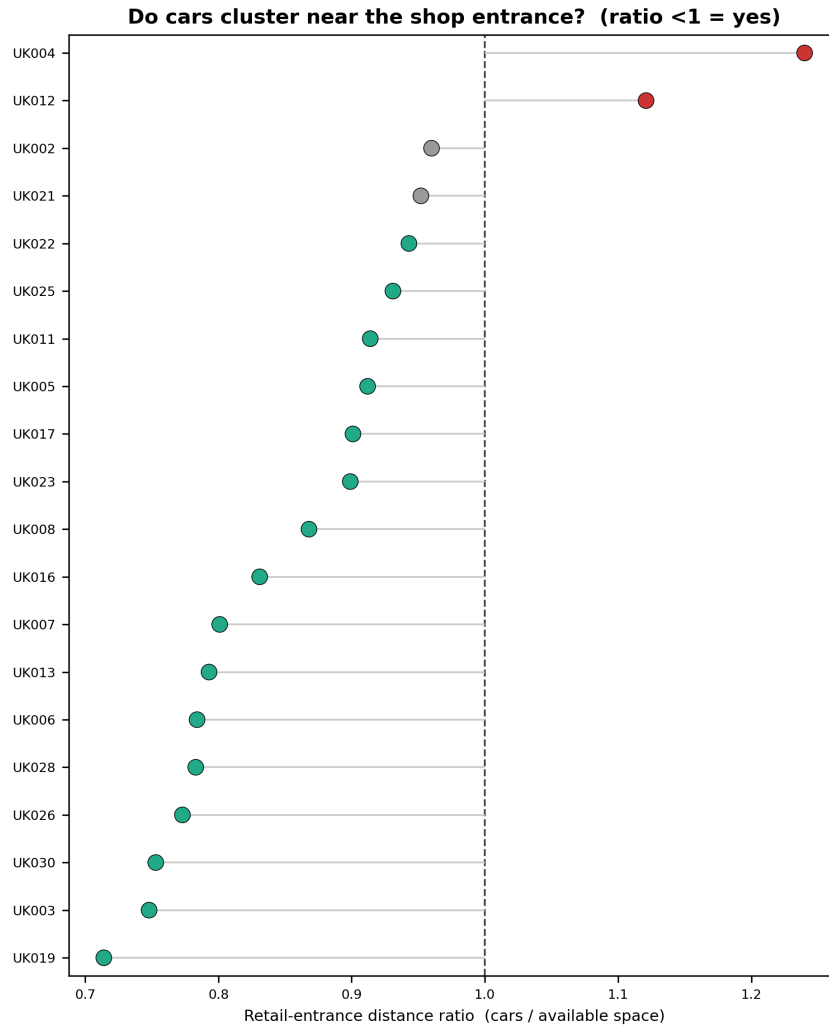


Figure 19: Retail-entrance distance ratio r_{dist} for each of the twenty held-out test sites, ordered. Values left of the dashed parity line ($r_{\text{dist}} = 1$) indicate vehicles closer to the store entrance than the average available parking space; eighteen of twenty sites fall on this side (green). Only UK012 and UK004 (red) park away from the entrance.

Parking fill vs distance to the retail entrance, representative sites

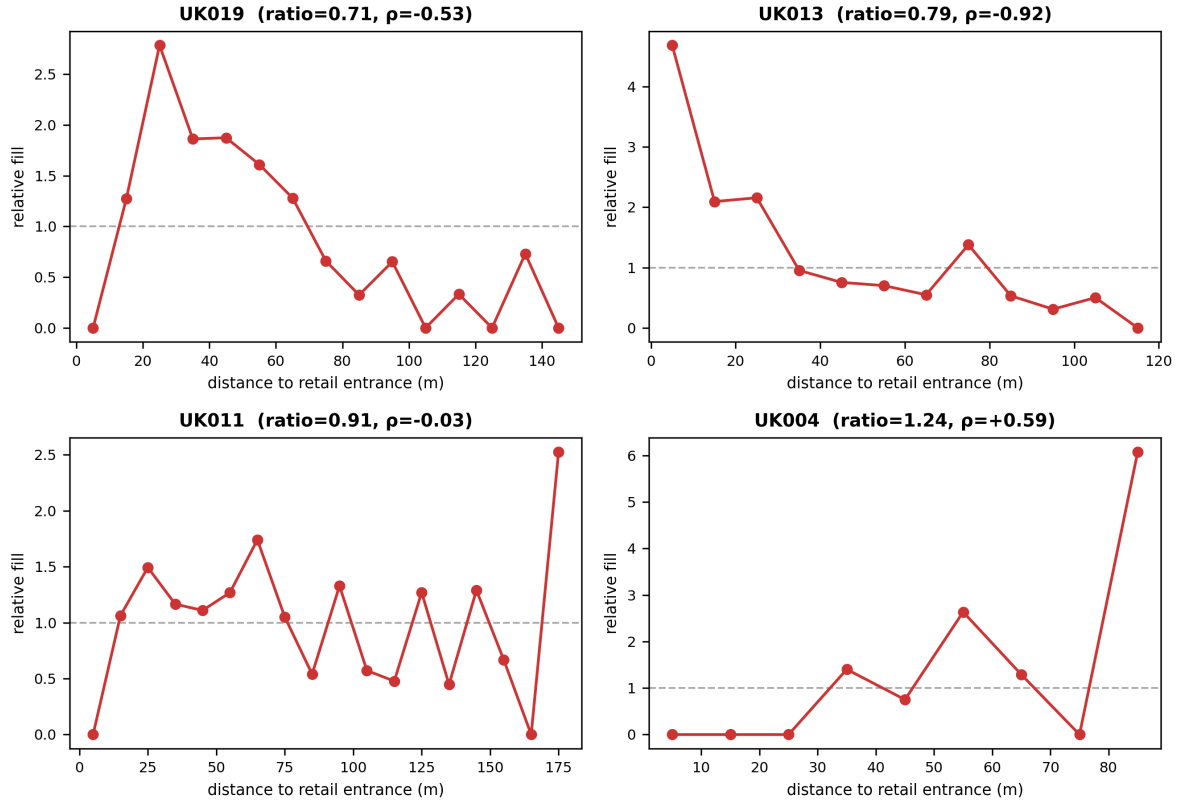


Figure 20: Relative fill (share of vehicles divided by share of available space) against distance to the retail entrance, for four representative sites spanning the range of behaviour, from the strongest entrance clustering (UK019, UK013) through a near-flat lot (UK011) to the away-from-entrance exception (UK004). A curve declining below the parity line with distance is the signature of a lot that fills from the entrance outward.

Relative to the *car-park* (vehicle) entrance, no comparable pattern is present. The distance ratios cluster around unity (mean 0.97), only twelve of the twenty sites fall below one, and the gradient averages effectively zero ($\rho_s = +0.02$, negative on nine of twenty sites). Where a lot does fill toward its vehicle entrance it is largely because that entrance happens to lie near the store. Taken across sites, parking demand shows no systematic relationship to the point at which vehicles enter the lot. The contrast between the two entrance types is the substantive result, and its interpretation, that within-lot demand is organised around the pedestrian destination rather than the vehicular origin, is developed in Section 5.4.

This accords with long-standing findings in transport research that proximity to the trip destination, rather than to the point of vehicular entry, is a primary determinant of parking choice (Axhausen & Polak 1991, Bonsall & Palmer 2004). What the pipeline adds is that this behavioural regularity can be recovered directly from overhead imagery, resolved at the level of the individual lot.

4.6 Summary of findings

On detection, the source models reproduced the reference within-domain accuracies (0.934 mAP@0.5 on COWC, 0.818 on VEDAI) but retained only about a fifth of that accuracy across domains, and pooling the two sources restored it (0.943 on COWC, 0.696 on VEDAI). On the twenty held-out UK test sites, mAP@0.5 rose at each stage of adaptation: 0.267 for the stock baseline, 0.528 for the zero-shot transfer model, and 0.886 for the UK-adapted model.

On the applied task, only the adapted model produced usable counts: a mean absolute error of 5.4 cars per site and occupancy recovered to within 2.6 percentage points of capacity on average ($R^2 = 0.97$ for both), whereas the baseline and transfer models returned negative R^2 on count and occupancy alike. Spatially, on eighteen of the twenty sites vehicles sat closer to the retail entrance than the average available parking space (mean distance ratio 0.88, mean $\rho_s = -0.30$), while no comparable relationship held for the vehicle entrance (mean 0.97, $\rho_s \approx 0$).

5 Discussion

5.1 Detection versus localisation, and the limits of square boxes

Across every model the detection metric (mAP@0.5) far exceeds the strict localisation metric (mAP@0.5:0.95). The detectors find vehicles far more reliably than they place tight boxes on them (Table 6). There are two factors that account for this. The first is intrinsic to the task. At 25 cm a car spans only 14–18 pixels, so a two- or three-pixel error in a box edge changes the IoU substantially. The strict metric, which raises the overlap requirement as high as 0.95, is therefore hard for objects this small regardless of domain (Tong et al. 2020). The in-domain source models show the same pattern, around 0.93 at an IoU of 0.5 but only about 0.49 over the strict range (Table 5). This confirms that it is a property of small-object detection rather than an artefact of the UK adaptation. The second factor is specific to this project’s design and was anticipated in the methodology. The public training data carries fixed-size square boxes (Figure 8), a form dictated by COWC providing only centre points, whereas the UK ground truth carries tight axis-aligned boxes drawn to each vehicle (Section 3.7). A detector trained to emit approximately square boxes is therefore scored against tight boxes of varying size and aspect ratio, which depresses the strict-IoU metric further. The reported mAP@0.5:0.95 is, in this respect, a *conservative* estimate of the square-trained models’ localisation ability rather than an inflated one. Consistent with this reading, fine-tuning improves the strict metric proportionally more than the loose one: from transfer to UK-adaptation, mAP@0.5 rises by a factor of 1.7 (0.528 to 0.886) but mAP@0.5:0.95 by a factor of 2.4 (0.146 to 0.352). As the UK calibration boxes are themselves tight, part of the adaptation gain is the model learning the target box *geometry*, not merely the target appearance. For the present purpose this ordering is reassuring rather than limiting. Vehicle counting relies on *detection* rather than precise footprint recovery, making the adapted model’s mAP@0.5 of 0.886 the operative metric. The strict-IoU shortfall merely reflects loose localisation, a deliberate and acknowledged trade-off of using square boxes. It provides clear motivation for the oriented-bounding-box extension proposed in future work: replacing the fixed square with a rotated box trained on an aerial benchmark such as DOTA (Xia et al. 2018) should recover the tight overlap that strict-IoU metrics demand while resolving the train/test geometry mismatch entirely.

5.2 The domain gap and the value of in-domain calibration

The single most useful result of the project is not the final accuracy but the shape of the path to it. The three UK experiments form a ladder, from 0.267 to 0.528 and finally 0.886 mAP@0.5 (Section 4.2). The domain-gap view (Section 4.3) shows the same detector losing two-fifths of its accuracy as the evaluation domain moves from its own source to UK retail. More interesting is where that accuracy comes from: fine-tuning on just eight UK sites added 0.36 mAP@0.5 , more than the 0.26 that the entire overhead-vehicle pretraining

stage added over the stock baseline. A small, carefully chosen quantity of in-domain data was worth more than a large volume of related but out-of-domain data. This is exactly the behaviour the transfer-learning framework of Section 2.6 predicts (Weiss et al. 2016), and the two stages play distinct, complementary roles. Pretraining on overhead vehicles buys *precision*: it teaches the detector what a car looks like from above and so suppresses the false positives that dominate the stock baseline, but it leaves recall capped near 0.59. In other words, more than 40% of UK vehicles are never even proposed as cars because the source distribution differs from UK retail in appearance, layout and density. Fine-tuning buys the missing *recall*: a small amount of UK data raises the ceiling from 0.59 to 0.94, making detectable the vehicles the foreign model could not see, while retaining the precision pretraining had established. Neither stage alone is sufficient, it is their order and combination that reaches a deployable detector, and the combined two-source model was itself necessary, since pooling COWC and VEDAI was what recovered the cross-domain coverage that either specialist lacked (Section 4.1). For an applied deployment this is the operative finding. The target domain has no public labelled data, but reaching it turned out to be inexpensive, only eight hand-labelled sites converted an out-of-domain detector into a near-in-domain one. A commercial user therefore does not need a large UK dataset to obtain accurate counts, only a small calibration set constructed with care. The qualification is that “care” is doing real work here. The eight sites were chosen to span retailers and lot sizes, and the balance-aware split mitigates but does not eliminate the risk that a detector calibrated on this spread would degrade on layouts or chains outside it. The strength of the result is its cost-effectiveness. Its boundary is the breadth of the calibration set.

5.3 Workability of the occupancy estimate

The counting stage answers the question the project set out to test: from a single overhead image, can a car park’s occupancy be estimated accurately? On the twenty held-out sites the adapted detector recovered occupancy to within 2.6 percentage points of capacity, with predicted and true counts lying along the identity line at $R^2 = 0.97$ (Section 4.4). At single-site level this is a workable estimate. It is accurate enough to distinguish a quiet lot from a busy one and to rank sites by how full they are, which is the immediate applied use.

One clarification governs how the estimate is read, occupancy here is a *utilisation* measure of detected cars over the counted bays, and thus can exceed 100% where vehicles park in aisles or loading areas, rather than a strict fraction of marked bays. Beyond that, what has been measured is occupancy at a single instant, not activity over time. Due to the fact that each site was observed in one Digimap capture, the project yields a snapshot per store, not the temporal “activity” signal. That is the rise and fall of trade across hours, days and seasons (Katona et al. 2025) that the proxy ultimately targets. This is a

matter of data rather than method. The same detect–count–occupancy pipeline applied to repeated captures would produce that series directly, so the project settles the harder, per-capture half of the problem to a deployable standard and leaves the temporal half as a data-acquisition step. The denominator’s own uncertainty and the single-capture constraint are set out as limitations in Section 5.5.

Underlying the whole claim is the methodological correction that distinguishes this work from its closest precedent (Leblanc et al. 2021). By reporting detection quality and counting quality separately, the project shows that a respectable detection score does not guarantee a usable count. The transfer model scored 0.528 mAP@0.5 yet produced counts with negative R^2 . Only the domain-adapted model reached a standard at which the occupancy estimate can be trusted, and it is that separation (refusing to infer counting accuracy from detection accuracy) that makes the occupancy claim defensible rather than optimistic.

5.4 From counting to geography

The counting stage collapses each site to a single number, but the detections that produce it are individually located, so re-expressing them as spatial data turns the occupancy estimate into a map of *where* demand falls. Each detected vehicle box was reduced to its centroid and projected into British National Grid coordinates (EPSG:27700) using the clip’s geo-referencing, yielding for every site a set of georeferenced car points and a Gaussian kernel-density surface over them (Silverman 1986) (Figure 21). The density surface is not scored against ground truth; it shows, qualitatively, what the validated counts of Section 4.4 make possible.

Read across the twenty test sites, three features are consistent. First, parking is strongly clustered rather than uniform, so demand concentrates into particular zones that the aggregate count cannot reveal, and two lots with similar totals can have very different internal structure. Second, that structure tracks lot geometry: long and narrow lots fill along a single central spine (Figure 21a), compact lots concentrate centrally, and multi-section lots show a separate hotspot per section. Third, and most useful for the occupancy question, partial occupancy is spatially explicit. On a large, roughly one-third-full lot the vehicles pack into one region while marked bays elsewhere sit visibly empty (Figure 21b). Occupancy, in other words, has a location: a site is not merely “36% full” but full *here* and empty *there*, information a single ratio discards.

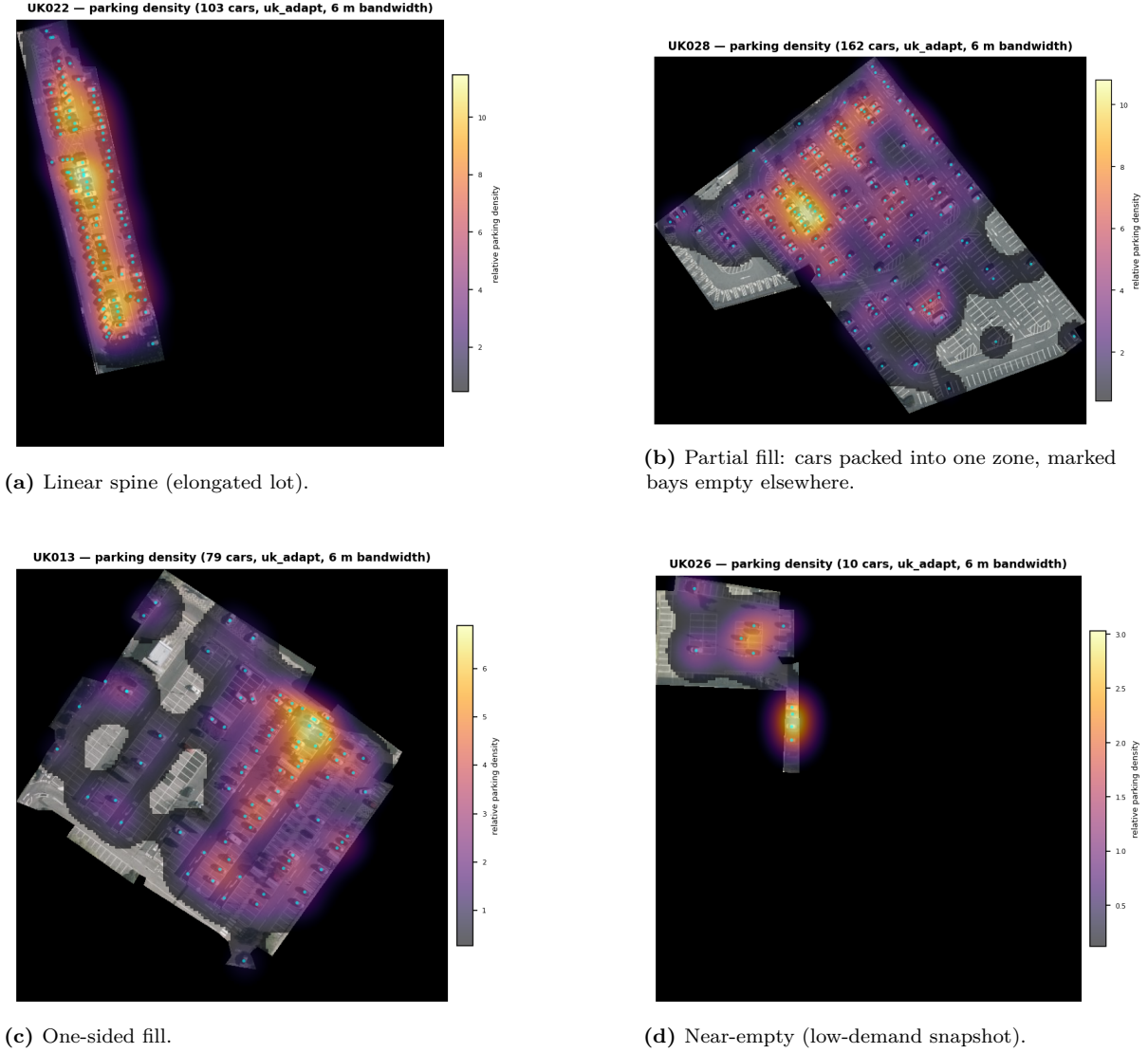


Figure 21: Kernel-density surfaces of parking demand for four contrasting held-out sites, built from the adapted detector’s georeferenced car centroids (cyan points) with a 6 m bandwidth. The maps reveal within-lot structure that the aggregate count cannot: a demand **(a)** spine in an elongated lot, **(b)** a large lot only partially filled with visibly empty bays, **(c)** one-sided fill, and **(d)** a near-empty low-demand lot. The colour scale is normalised *within* each site, so intensity is comparable between rows of one lot but not between panels; only the car count is comparable across sites.

The observation that lots fill from their most convenient bays can be made precise. The entrance-proximity analysis of Section 4.5 replaces the visual impression of the density maps with a measured relationship, comparing where vehicles sit against where the lot’s available space sits, relative to two digitised points of access: the store’s pedestrian entrance and the vehicle entrance to the lot. The two behave very differently, and the contrast is the substantive spatial finding. Relative to the *retail* entrance, demand is consistently drawn toward the store. On eighteen of the twenty sites, vehicles lie closer to the entrance than the average parking space does (mean distance ratio 0.88), and relative fill declines with distance from it (mean $\rho_s = -0.30$). Relative to the *vehicle* entrance there is no such pattern (mean ratio 0.97, $\rho_s \approx 0$); where a lot does happen to fill toward

its vehicle entrance, it is largely because that entrance lies near the store.

The interpretation is straightforward and, importantly, consistent across retailers and lot geometries. Within-lot demand is organised around the *pedestrian destination*, not the *vehicular origin*. Drivers do not settle near where they enter the lot, instead they press toward where they are walking to, so the bays nearest the storefront fill first and the lot empties outward from the door. This is the behavioural regularity that the density surfaces only hinted at, now quantified, and it strengthens the footfall-proxy premise on which the whole project rests (Section 2.8). If parking concentrates predictably around the entrance, the near-entrance region is the most informative place to observe demand, and a partially full lot is not spatially ambiguous but structured. It also gives the aggregate occupancy figure a mechanism, explaining *why* the empty bays of Figure 21b sit where they do.

Two sites run counter to the pattern and bound the claim. UK004 carries only eight detected vehicles, too few for its distance ratio to be meaningful, and is better set aside than interpreted. UK012 is a genuine exception in which demand sits away from the digitised entrance. This most likely stems from structural factors rather than unexpected driver behaviour. For instance, the store may possess multiple entrances despite only the primary access point being digitised. Alternatively, the bays closest to the frontage may be reserved for trolley parks, accessible parking, or loading lanes, rendering them unavailable to the general parking demand measured here. Representing each site by a single, manually designated entrance is a deliberate methodological simplification, and UK012 highlights precisely where that assumption reaches its limit.

The entrance measure carries its own simplifications. A single hand-placed entrance per type, straight-line rather than walking distance, and an area- rather than bay-level denominator, which, together with the density surface's caveats, are catalogued in Section 5.5. What remains substantively is the temporal dimension, already identified in Section 5.3 as the primary boundary of the occupancy claim. Since the spatial and entrance-aware evaluation layers are built directly on top of the underlying vehicle counts, they are fully prepared to process time-series data without modification as soon as repeated imagery becomes available. That is the core contribution of this stage, a detection pipeline whose output is not merely counted, but geographically mapped with a measured behavioural structure.

5.5 Limitations

Several limitations bound the results and set the agenda for the future work that follows.

- **Loose localisation.** The detector was trained on fixed-size square boxes synthesised for COWC but evaluated against tight UK boxes, so its strict-IoU localisation is weak

(mAP@0.5:0.95 of 0.352) even though its detection and counting are strong (Section 5.1). This is acceptable for counting, which depends only on finding each vehicle, but it limits any downstream use that needs an accurate vehicle footprint, and it is the clearest target for the oriented-bounding-box extension.

- **A single capture per site.** The imagery gives one snapshot per store, so the occupancy estimate is static and the inherently temporal footfall-proxy premise cannot be validated here (Section 5.3).
- **Capacity-count error.** Each site’s capacity was obtained by manually counting its marked bays from the imagery. That is a direct count, but not error-free where markings are faded or occluded, or where overflow parking blurs the true bay count. This uncertainty enters the occupancy rate but not the vehicle count (Section 5.3).
- **Small calibration and threshold sets.** Adaptation used eight sites and the operating threshold was fixed on only two, which is enough to demonstrate the effect but thin for guaranteeing generalisation. A wider calibration set spanning more retailers, regions and lot geometries would strengthen the claim and stabilise the threshold.
- **Non-UK, square-box source data.** The public sources are US (COWC) and largely European-rural (VEDAI). The ideal addition, ITCVD (Yang et al. 2019), a European, nadir, tight-box dataset, could not be obtained during the project. DOTA (Xia et al. 2018), though large and tightly annotated, is unsuitable as a drop-in source because its ground sampling distance varies per image and its licence restricts use to academic research.
- **The hardest lots remain hardest.** Residual counting error concentrates on the densest, most occluded car parks, where recall dips and the strict-IoU penalty is greatest; a single architecture (YOLOv8s) was used throughout, and the small-object architectural modifications discussed in Section 5.1 were not evaluated, so there is likely headroom on exactly these sites.
- **Idealised entrances in the spatial analysis.** The entrance-proximity result (Section 5.4) represents each site by a single hand-digitised pedestrian entrance and a single vehicle entrance, measures straight-line rather than walking distance, and scores vehicle positions against the lot’s available *area* rather than its marked bays. These simplifications are adequate for the cross-site regularity reported here but not for a per-bay account of occupancy, and the single-entrance idealisation is the likely source of the one clear exception (UK012). Multiple digitised entrances, network distances along drivable and walkable paths, and a bay-level rather than area-level denominator would each sharpen the measure.

6 Conclusion and future directions

6.1 Summary of contributions

This dissertation set out to test whether vehicles in dense UK retail car parks can be detected and counted from approximately 25 cm aerial imagery accurately enough to estimate a site’s occupancy from a single image, and to establish what that requires. The answer is affirmative, subject to two conditions the project makes explicit: imagery of sufficient resolution, and a detector adapted to the target domain. The scope of this result should be stated precisely. What the work directly demonstrates is single-image vehicle detection and occupancy estimation on held-out UK sites. Using those occupancy estimates as a measure of retail demand or trading activity is the premise that motivates the work, not a claim it tests here, because that would require repeated observations validated against an independent measure of activity (Section 6.3). Within this scope the central argument is borne out. The barrier to estimating UK car-park occupancy from above has been one of *resolution and evaluation* rather than of method. The contributions that support this conclusion are methodological, empirical, and practical in turn.

The principal *methodological* contribution is the separation of detection from counting. Prior UK work conflated the two, reporting a single count-regression statistic from which detection quality could not be recovered (Section 2.8). This project instead reports detection metrics and counting error as distinct quantities, so that a localisation failure can never be mistaken for a counting success, nor an accurate total credited to a detector whose errors merely cancel. Around this correction sits a resolution-first pipeline: public overhead-vehicle sources are resampled to the 25 cm resolution of the target before any adaptation, converting the downsampling of COWC’s 15 cm imagery into a lightweight domain-alignment step (Section 2.6), and the whole is evaluated on a frozen, balance-aware, site-level split that prevents overlapping tiles from leaking cars across the train and test boundary. A final methodological contribution is the entrance-proximity framework itself, which re-expresses located detections as georeferenced points and summarises each lot with two interpretable statistics, a distance ratio and a Spearman gradient (Section 4.5), turning a per-site total into a measure of where within the lot demand falls.

The *empirical* contributions quantify what each of these design choices buys. Detection accuracy rose at each stage of adaptation, from 0.267 mAP@0.5 for the stock baseline to 0.528 for the zero-shot transfer model and 0.886 for the UK-adapted model (Section 4.2). Placing the source and UK results on a single axis measured the domain gap directly: the same detector falls by 0.41 mAP@0.5 from its in-domain COWC score to zero-shot UK, of which fine-tuning on eight hand-labelled sites recovers 0.36 (Section 4.3). The most striking feature of that ladder is that these eight sites contributed more than the entire overhead-vehicle pretraining stage, so a small, carefully chosen quantity of in-domain data

outweighed a large volume of related but out-of-domain data (Section 5.2). Carried into counting, only the adapted detector produced usable estimates, recovering site occupancy to within 2.6 percentage points of capacity at $R^2 = 0.97$, while the baseline and transfer models returned negative R^2 (Section 4.4). That last contrast is itself an empirical result, and the clearest vindication of the methodological separation above. The transfer model is a competent detector at 0.528 mAP@0.5 yet cannot count, because its false positives and missed vehicles do not cancel. Finally, because every detection is located, the counts were re-expressed as within-lot geography, yielding a consistent spatial pattern. On eighteen of the twenty test sites parked vehicles concentrate around the retail (pedestrian) entrance rather than the vehicular one, with no comparable pull toward the point of vehicular entry (Section 5.4). This is a single-snapshot spatial measurement. Its reading as customer behaviour, though consistent with established parking research, is interpretive rather than something the single-capture design establishes on its own.

The *practical* contribution is a working, validated recipe for a setting that previously had none. For a target domain with no public labelled data, the project shows that a modest, carefully constructed calibration set, here just eight sites, is both necessary and, on this evidence, largely sufficient to reach near-in-domain performance, which lowers the cost of standing up the same pipeline on a new retailer, region or country to a small labelling exercise rather than a bespoke data-collection programme. It also settles a concrete data-sourcing question: imagery at around 25 cm suffices where the metre-scale satellite imagery of earlier UK work does not, so the resolution of the source, not the sophistication of the detector, is where practitioners should spend first. And by delivering the spatial distribution of parking as a map rather than a single scalar, the pipeline offers a within-lot view of where vehicles concentrate that an aggregate occupancy figure discards. Whether that concentration can be read as a fine-grained signal of customer demand is a promising but untested extension. The pipeline establishes the measurement, not yet its economic interpretation.

6.2 Strengths and weaknesses

The method’s principal strength is the combination of a resolution-first design with a disciplined evaluation. Choosing 25 cm imagery placed the task inside the stable, high-accuracy region of the accuracy–resolution curve, rather than past the cliff where earlier UK work operated, and reporting detection and counting separately ensured that a good count could never be mistaken for accurate detection. The frozen, balance-aware, site-level split gives the headline numbers an honest, leak-free basis, and the whole result is cost-effective. Eight labelled sites converted an out-of-domain detector into a near-in-domain one. A further strength is that the georeferenced output turns each count into a map, enabling the entrance-proximity analysis and the behavioural finding that a single occupancy ratio discards.

Its weaknesses follow from the same design choices and are set out in full in Section 5.5. Localisation is loose, a consequence of training on fixed-size square boxes and scoring against tight ones. The estimate is a single-instant snapshot, so the inherently temporal footfall-proxy premise cannot be validated here. Additionally, the calibration and threshold sets are thin. A limitation of the training datasets is that the public sources are non-UK and square-box. Residual counting error concentrates on the densest, most occluded lots and the spatial measure idealises each site to a single hand-placed entrance. None of these undermines the counting result, which is verified against hand annotations, but each bounds how far the claims extend.

6.3 Further challenges and future directions

The clearest next step is temporal. Each site was observed only once, this project resolves the primary challenge of a single-capture detection to a deployable standard, while leaving the temporal activity signal unmeasured. Applying this exact detect–count–occupancy pipeline to repeated captures of the same sites would turn each static estimate, and each entrance gradient, into the time series the proxy ultimately targets (Section 5.3). This expansion represents a data-acquisition step rather than a methodological one. The existing pipeline is fully prepared to process time-series data without modification.

The most concrete architectural extension concerns the bounding-box geometry itself. The fixed-size square adopted for COWC (Figure 8) is the correct orientation-agnostic choice given that dataset’s centre-point labels, but the box-size analysis exposed its intrinsic limit. On densely parked, angled rows a square sized to the car’s longest dimension necessarily over-covers the short axis, so adjacent boxes clip into one another. This is not a tuning error, but a property of placing axis-aligned boxes on rotated, tightly packed objects. Indeed, the empirical analysis confirmed that the 14 px is the optimal footprint, with 16 px over-merging and 12 px under-covering. Transitioning to oriented bounding boxes (OBB) involves more than simply flipping a switch, owing to two main constraints. COWC provides no per-car orientation, so a correct rectangle cannot be derived, and YOLOv8 predicts axis-aligned boxes only. The natural pilot is therefore to trial an oriented-box detector, such as YOLOv11’s OBB head (Murat & Kiran 2025) on VEDAI, where per-vehicle corner coordinates (Razakarivony & Jurie 2016) enable oriented ground truth to be constructed directly. Comparing this model against the YOLOv8 square-box baseline on the same VEDAI subset would demonstrate whether oriented boxes reduce overlap in dense, angled rows and improve the strict-IoU localization that square boxes depress (Section 5.1).

The tighter, detail-preserving boxes produced by an OBB model align with the small-object evidence of Li et al. (2024), with DOTA (Xia et al. 2018) serving as a primary dataset for scaling the experiment up. This direction is framed as future work because it requires

both a specialized detector architecture and orientation labels absent from the primary training source. Nevertheless, a single, tightly scoped experiment would be sufficient to convert this structural proposal into empirical evidence.

Two further directions would refine the underlying methodology. Firstly, a wider calibration set spanning more retailers, regions and lot geometries would stabilise the operating threshold and strengthen the generalisation claim. Secondly, the entrance analysis would be sharpened by digitising multiple entrances per site, measuring network (walking) distances rather than straight lines, and adopting a bay-level rather than area-level denominator.

References

- Akyon, F. C., Altinuc, S. O. & Temizel, A. (2022), Slicing aided hyper inference and fine-tuning for small object detection, *in* ‘IEEE International Conference on Image Processing (ICIP)’, pp. 966–970.
- Axhausen, K. W. & Polak, J. W. (1991), ‘Choice of parking: Stated preference approach’, *Transportation* **18**(1), 59–81.
- Bonsall, P. & Palmer, I. (2004), ‘Modelling drivers’ car parking behaviour using data from a travel choice simulator’, *Transportation Research Part C: Emerging Technologies* **12**(5), 321–347.
- Cortes, C. & Vapnik, V. (1995), ‘Support-vector networks’, *Machine Learning* **20**(3), 273–297.
- CVAT.ai Corporation (2026), ‘Computer vision annotation tool (CVAT)’, Self-hosted via Docker from <https://github.com/cvat-ai/cvat>. Commit <hash>.
- Dalal, N. & Triggs, B. (2005), Histograms of oriented gradients for human detection, *in* ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 886–893.
- Di Mauro, D., Furnari, A., Patanè, G., Battiato, S. & Farinella, G. M. (2019), ‘Estimating the occupancy status of parking areas by counting cars and non-empty stalls’, *Journal of Visual Communication and Image Representation* **62**, 234–244.
- Donaldson, D. & Storeygard, A. (2016), ‘The view from above: Applications of satellite data in economics’, *Journal of Economic Perspectives* **30**(4), 171–198.
- Everingham, M., Van Gool, L., Williams, C. K. I., Winn, J. & Zisserman, A. (2010), ‘The PASCAL visual object classes (VOC) challenge’, *International Journal of Computer Vision* **88**(2), 303–338.
- Fang, X., Jeon, M., Qin, Z., Panev, S., de Melo, C., Hu, S., Chakraborty, S. & De la Torre, F. (2025), Adapting vehicle detectors for aerial imagery to unseen domains with weak supervision, *in* ‘Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)’.
- Felzenszwalb, P. F., Girshick, R. B., McAllester, D. & Ramanan, D. (2010), ‘Object detection with discriminatively trained part-based models’, *IEEE Transactions on Pattern Analysis and Machine Intelligence* **32**(9), 1627–1645.
- Feng, S. & Fay, S. (2022), ‘An empirical investigation of forward-looking retailer performance using parking lot traffic data derived from satellite imagery’, *Journal of Retailing* **98**(4), 633–646.

- Ge, Z., Liu, S., Wang, F., Li, Z. & Sun, J. (2021), ‘YOLOX: Exceeding YOLO series in 2021’, *arXiv preprint arXiv:2107.08430*.
- GEOLYTIX (2024), ‘Supermarket retail points’, Open data, <https://geolytix.com/blog/supermarket-retail-points/>. Insert the exact version (e.g. v31) and access date used for site selection.
- Getmapping Plc (2021), ‘High resolution (25 cm) vertical aerial imagery [tiff geospatial data]’, EDINA Aerial Digimap Service. Downloaded via UCL licence; use the exact tiles/date from the Digimap-generated citation.
- Girshick, R., Donahue, J., Darrell, T. & Malik, J. (2014), Rich feature hierarchies for accurate object detection and semantic segmentation, *in* ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 580–587.
- Gliaubičiūtė, D., Janavičius, R., Gadeikytė, A. & Paulauskas, L. (2023), Influence of aerial image resolution on vehicle detection accuracy, *in* ‘Proceedings of the 28th International Conference on Information Society and University Studies (IVUS 2023)’, Vol. 3575 of *CEUR Workshop Proceedings*.
- González, M. C., Hidalgo, C. A. & Barabási, A.-L. (2008), ‘Understanding individual human mobility patterns’, *Nature* **453**(7196), 779–782.
- He, K., Zhang, X., Ren, S. & Sun, J. (2016), Deep residual learning for image recognition, *in* ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 770–778.
- Hellekes, J., Kehlbacher, A., López Díaz, M., Merkle, N., Henry, C., Kurz, F. & Heinrichs, M. (2023), ‘Parking space inventory from above: Detection on aerial images and estimation for unobserved regions’, *IET Intelligent Transport Systems* **17**(5), 1009–1021.
- Henderson, J. V., Storeygard, A. & Weil, D. N. (2012), ‘Measuring economic growth from outer space’, *American Economic Review* **102**(2), 994–1028.
- Hsieh, M.-R., Lin, Y.-L. & Hsu, W. H. (2017), Drone-based object counting by spatially regularized regional proposal network, *in* ‘Proceedings of the IEEE International Conference on Computer Vision (ICCV)’, pp. 4145–4153.
- Jean, N., Burke, M., Xie, M., Davis, W. M., Lobell, D. B. & Ermon, S. (2016), ‘Combining satellite imagery and machine learning to predict poverty’, *Science* **353**(6301), 790–794.
- Jocher, G., Chaurasia, A. & Qiu, J. (2023), ‘Ultralytics YOLO’, <https://github.com/ultralytics/ultralytics>. Software; set exact version from training logs.
- Katona, Z., Painter, M., Patatoukas, P. N. & Zeng, J. (2025), ‘On the capital market conse-

- quences of big data: Evidence from outer space’, *Journal of Financial and Quantitative Analysis* **60**(2), 551–579.
- Krizhevsky, A., Sutskever, I. & Hinton, G. E. (2012), ImageNet classification with deep convolutional neural networks, in ‘Advances in Neural Information Processing Systems (NeurIPS)’, Vol. 25, pp. 1097–1105.
- Leblanc, M., Van Dijk, J., Rains, T. & Jack, H. (2021), ‘Predicting car park occupancy rates from satellite images with a deep learning model’, UCL Consumer Data Research Centre (CDRC), Masters Dissertation Scheme project (with Sainsbury’s).
- LeCun, Y., Bengio, Y. & Hinton, G. (2015), ‘Deep learning’, *Nature* **521**(7553), 436–444.
- LeCun, Y., Bottou, L., Bengio, Y. & Haffner, P. (1998), ‘Gradient-based learning applied to document recognition’, *Proceedings of the IEEE* **86**(11), 2278–2324.
- Lempitsky, V. & Zisserman, A. (2010), Learning to count objects in images, in ‘Advances in Neural Information Processing Systems (NeurIPS)’, Vol. 23.
- Li, K., Wan, G., Cheng, G., Meng, L. & Han, J. (2020), ‘Object detection in optical remote sensing images: A survey and a new benchmark’, *ISPRS Journal of Photogrammetry and Remote Sensing* **159**, 296–307.
- Li, Y., Li, Q., Pan, J., Zhou, Y., Zhu, H., Wei, H. & Liu, C. (2024), ‘SOD-YOLO: A small-object-detection algorithm based on improved YOLOv8 for UAV images’, *Remote Sensing* **16**(16), 3057.
- Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B. & Belongie, S. (2017), Feature pyramid networks for object detection, in ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 2117–2125.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P. & Zitnick, C. L. (2014), Microsoft COCO: Common objects in context, in ‘Computer Vision – ECCV 2014’, Vol. 8693 of *Lecture Notes in Computer Science*, Springer, pp. 740–755.
- Lipton, Z. C., Elkan, C. & Naryanaswamy, B. (2014), Optimal thresholding of classifiers to maximize F1 measure, in ‘Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2014)’, Vol. 8725 of *Lecture Notes in Computer Science*, Springer, pp. 225–239.
- Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y. & Berg, A. C. (2016), SSD: Single shot MultiBox detector, in ‘Computer Vision – ECCV 2016’, Vol. 9905 of *Lecture Notes in Computer Science*, Springer, pp. 21–37.
- Loshchilov, I. & Hutter, F. (2019), Decoupled weight decay regularization, in ‘International Conference on Learning Representations (ICLR)’.

- Lu, S., Fang, Z., Zhang, X., Shaw, S.-L., Yin, L., Zhao, Z. & Yang, X. (2017), ‘Understanding the representativeness of mobile phone location data in characterizing human mobility indicators’, *ISPRS International Journal of Geo-Information* **6**(1), 7.
- Mundhenk, T. N., Konjevod, G., Sakla, W. A. & Boakye, K. (2016), A large contextual dataset for classification, detection and counting of cars with deep learning, in ‘Computer Vision – ECCV 2016, Part III’, Vol. 9907 of *Lecture Notes in Computer Science*, Springer, pp. 785–800.
- Murat, A. A. & Kiran, M. S. (2025), ‘A comprehensive review on YOLO versions for object detection’, *Engineering Science and Technology, an International Journal* **70**, 102161.
- Neubeck, A. & Van Gool, L. (2006), Efficient non-maximum suppression, in ‘18th International Conference on Pattern Recognition (ICPR)’, pp. 850–855.
- Ojala, T., Pietikäinen, M. & Mäenpää, T. (2002), ‘Multiresolution gray-scale and rotation invariant texture classification with local binary patterns’, *IEEE Transactions on Pattern Analysis and Machine Intelligence* **24**(7), 971–987.
- Padilla, R., Netto, S. L. & da Silva, E. A. B. (2020), A survey on performance metrics for object-detection algorithms, in ‘27th International Conference on Systems, Signals and Image Processing (IWSSIP)’, pp. 237–242.
- Pan, S. J. & Yang, Q. (2010), ‘A survey on transfer learning’, *IEEE Transactions on Knowledge and Data Engineering* **22**(10), 1345–1359.
- QGIS Development Team (2026), ‘QGIS geographic information system’, <https://www.qgis.org>. Version 4.2.1.
- Razakarivony, S. & Jurie, F. (2016), ‘Vehicle detection in aerial imagery: A small target detection benchmark’, *Journal of Visual Communication and Image Representation* **34**, 187–203.
- Redmon, J., Divvala, S., Girshick, R. & Farhadi, A. (2016), You only look once: Unified, real-time object detection, in ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 779–788.
- Redmon, J. & Farhadi, A. (2018), ‘YOLOv3: An incremental improvement’, *arXiv preprint arXiv:1804.02767*.
- Ren, S., He, K., Girshick, R. & Sun, J. (2015), Faster R-CNN: Towards real-time object detection with region proposal networks, in ‘Advances in Neural Information Processing Systems (NeurIPS)’, Vol. 28.
- Shermeyer, J. & Van Etten, A. (2019), The effects of super-resolution on object detection

- performance in satellite imagery, in ‘Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)’.
- Shreeram H, Nagendra B S, Sougandh Sunil & Adwitiya Mukhopadhyay (2025), Smart parking space availability detection using aerial images and YOLO-based deep learning models, in ‘2025 Third International Conference on Networks, Multimedia and Information Technology (NMITCON)’, IEEE, p. TODO: from IEEE Xplore.
- Silverman, B. W. (1986), *Density Estimation for Statistics and Data Analysis*, Chapman & Hall, London.
- Soundararaj, B., Cheshire, J. & Longley, P. (2020), ‘Estimating real-time high-street footfall from wi-fi probe requests’, *International Journal of Geographical Information Science* **34**(2), 325–343.
- Sutskever, I., Martens, J., Dahl, G. E. & Hinton, G. E. (2013), On the importance of initialization and momentum in deep learning, in ‘Proceedings of the 30th International Conference on Machine Learning (ICML)’, Vol. 28 of *PMLR*, pp. 1139–1147.
- Tian, Z., Shen, C., Chen, H. & He, T. (2019), FCOS: Fully convolutional one-stage object detection, in ‘Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)’, pp. 9627–9636.
- Tong, K., Wu, Y. & Zhou, F. (2020), ‘Recent advances in small object detection based on deep learning: A review’, *Image and Vision Computing* **97**, 103910.
- Tuia, D., Persello, C. & Bruzzone, L. (2016), ‘Domain adaptation for the classification of remote sensing data: An overview of recent advances’, *IEEE Geoscience and Remote Sensing Magazine* **4**(2), 41–57.
- Wang, C.-Y. & Liao, H.-Y. M. (2024), ‘YOLOv9: Learning what you want to learn using programmable gradient information’, *arXiv preprint arXiv:2402.13616*.
- Wang, M. & Deng, W. (2018), ‘Deep visual domain adaptation: A survey’, *Neurocomputing* **312**, 135–153.
- Weiss, K., Khoshgoftaar, T. M. & Wang, D. (2016), ‘A survey of transfer learning’, *Journal of Big Data* **3**(1), 9.
- Xia, G.-S., Bai, X., Ding, J., Zhu, Z., Belongie, S., Luo, J., Datcu, M., Pelillo, M. & Zhang, L. (2018), DOTA: A large-scale dataset for object detection in aerial images, in ‘Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)’, pp. 3974–3983.
- Yang, M. Y., Liao, W., Li, X., Cao, Y. & Rosenhahn, B. (2019), ‘Vehicle detection

- in aerial images’, *Photogrammetric Engineering & Remote Sensing* **85**(4), 297–304. arXiv:1801.07339.
- Yosinski, J., Clune, J., Bengio, Y. & Lipson, H. (2014), How transferable are features in deep neural networks?, in ‘Advances in Neural Information Processing Systems (NeurIPS)’, Vol. 27, pp. 3320–3328.
- Zhu, C. (2019), ‘Big data as a governance mechanism’, *The Review of Financial Studies* **32**(5), 2021–2061.
- Zhu, P., Wen, L., Du, D., Bian, X., Fan, H., Hu, Q. & Ling, H. (2021), ‘Detection and tracking meet drones challenge’, *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(11), 7380–7399.
- Zou, Z., Chen, K., Shi, Z., Guo, Y. & Ye, J. (2023), ‘Object detection in 20 years: A survey’, *Proceedings of the IEEE* **111**(3), 257–276.

Appendices

Appendix A: Representative code

The data-preparation, training and evaluation pipeline was implemented as small, individually tested Python components driven by configuration files. This appendix reproduces the algorithmic core of the steps described in Section 3; routine input/output, argument parsing and validation are omitted for brevity. The complete, version-controlled implementation is available in the project repository.

Listing 1: Centre point to fixed-size, tile-normalised YOLO box, clipped to the tile boundary (Section 3.3).

```

1 def point_to_yolo_box(x, y, tile_w, tile_h, box_px, class_id=0):
2     half = box_px / 2.0
3     # Pixel corners, clipped so no box spills past the tile edge.
4     x1, y1 = max(x - half, 0.0), max(y - half, 0.0)
5     x2, y2 = min(x + half, tile_w), min(y + half, tile_h)
6     # Recompute centre/size from the clipped corners, then normalise to the tile.
7     xc = ((x1 + x2) / 2.0) / tile_w
8     yc = ((y1 + y2) / 2.0) / tile_h
9     return [class_id, xc, yc, (x2 - x1) / tile_w, (y2 - y1) / tile_h]
```

Listing 2: Overlapping sliding-window tile origins with an edge-flush terminal tile, guaranteeing full boundary coverage (Section 3.3).

```

1 def tile_origins(extent, tile_size, stride):
2     if extent <= tile_size:
3         return [0]
4     origins = list(range(0, extent - tile_size + 1, stride))
5     last = extent - tile_size
6     if origins[-1] != last:
7         origins.append(last) # edge-flush tile covers the far edge
8     return origins
```

Listing 3: Explicit, named scene/site-level split – everything not named as val/test is train – so overlapping tiles never leak across the split (Section 3.3.4).

```

1 def assign_split(scene_name, val_scenes, test_scenes=None):
2     if scene_name in set(test_scenes or []):
3         return "test"
4     if scene_name in set(val_scenes):
5         return "val"
6     return "train"
```

Listing 4: Seam-aware de-duplication: boxes from *different* tiles overlapping at $\text{IoU} \geq \text{threshold}$ are the same physical vehicle and merged; same-tile overlaps are kept (Sections 3.7 and 3.10).

```

1 def dedup(boxes, iou_thr=0.5): # boxes: [(xyxy_global, tile_id), ...]
2     kept = []
```

```

3   for bb, tile in boxes:
4       if any(kt != tile and iou(bb, kb) >= iou_thr for kb, kt in kept):
5           continue # duplicate of a car already kept
6       kept.append((bb, tile))
7   return len(kept)

```

Listing 5: Selecting the deployment threshold as the confidence that maximises F_1 : a monotone sweep over the score-sorted (score, is-TP) pairs (Sections 3.6 and 3.10).

```

1 def f1_optimal_threshold(scored, n_truth): # scored: [(score, is_tp), ...]
2     order = sorted(scored, key=lambda r: -r[0])
3     best, tp, fp = (0.0, 0.0), 0, 0 # (tau, f1)
4     for i, (score, is_tp) in enumerate(order):
5         tp += is_tp; fp += (1 - is_tp)
6         if i + 1 < len(order) and order[i + 1][0] == score:
7             continue # only test at a score boundary
8         p, r = tp / (tp + fp), tp / n_truth
9         f1 = 2 * p * r / (p + r) if (p + r) else 0.0
10        if f1 > best[1]:
11            best = (score, f1)
12    return best

```

Listing 6: Class-agnostic scoring of the stock baseline: all predicted COCO classes are collapsed to one foreground category, so any detection on a real car counts regardless of its COCO label (Section 3.9).

```

1 # The class name is relabelled for honest plots (display only, not scoring).
2 if single_cls:
3     label0 = yaml.safe_load(open(data_yaml))["names"][0]
4     model.model.names = {i: label0 for i in model.model.names}
5 metrics = model.val(data=data_yaml, split="test",
6                     conf=0.001, iou=0.6, single_cls=single_cls)

```

Listing 7: Georeferencing: recover the clip origin from tile_index.csv and map a global clip pixel to British National Grid coordinates (Section 3.11).

```

1 def load_geotransform(site_dir):
2     r = list(csv.DictReader(open(site_dir / "tile_index.csv")))[0]
3     px = float(r["pixel_size"])
4     origin_x = float(r["ulx"]) - int(r["col_off"]) * px
5     origin_y = float(r["uly"]) + int(r["row_off"]) * px
6     return origin_x, origin_y, px, r.get("crs", "EPSG:27700")
7
8 def pixel_to_ground(gx, gy, geo):
9     origin_x, origin_y, px, _ = geo
10    return origin_x + gx * px, origin_y - gy * px # rows increase southward

```


Listing 8: Entrance-proximity core: the distance ratio and the relative-fill gradient, comparing detected cars against grid-sampled available space (Section 3.12).

```

1 def entrance_stats(car_xy, area_xy, entrances, ring=10.0):
2     # nearest-entrance distance for cars and for available-space samples
3     d_car = nearest_dist(car_xy, entrances)
4     d_area = nearest_dist(area_xy, entrances)
5     # distance ratio: < 1 => cars nearer the entrance than the average space
6     r_dist = d_car.mean() / d_area.mean()
7     # relative fill per concentric ring, then Spearman(distance, fill)
8     edges = np.arange(0, max(d_car.max(), d_area.max()) + ring, ring)
9     mid = (edges[:-1] + edges[1:]) / 2.0
10    car_share = np.histogram(d_car, edges)[0] / len(d_car)
11    area_share = np.histogram(d_area, edges)[0] / len(d_area)
12    ok = area_share > 0
13    rel = car_share[ok] / area_share[ok] # relative fill (> 1 = fuller ring)
14    rho = spearman(mid[ok], rel) # < 0 = fills from entrance outward
15    return r_dist, rho

```

Appendix B: UK site registry

The thirty UK retail sites, their estimated capacities and size classes, and the frozen calibration/test split, read unchanged at every stage of the UK pipeline (Section 3.8). Capture dates are recorded so that each observation can be tied to a known acquisition.

Table 9: UK site registry (accepted sites). Split: Calib. = calibration pool (UK009 and UK020 serve as the internal monitoring set); Test = held-out evaluation.

Site	Retailer	Cap.	Size	Split	Capture date
UK001	Aldi	100	Medium	Calib.	2023-05-20
UK002	Aldi	180	Medium	Test	2021-04-24
UK003	Aldi	95	Medium	Test	2021-04-24
UK004	Aldi	130	Medium	Test	2021-07-17
UK005	Asda	260	Large	Test	2022-07-19
UK006	Asda	105	Medium	Test	2022-04-26
UK007	Iceland	250	Large	Test	2022-04-30
UK008	Lidl	130	Medium	Test	2021-04-23
UK009	Lidl	50	Small	Calib.	2023-05-21
UK010	Lidl	25	Small	Calib.	2021-04-24
UK011	Marks and Spencer	400	Large	Test	2022-04-30
UK012	Morrisons	140	Medium	Test	2023-07-03
UK013	Morrisons	100	Medium	Test	2022-04-30
UK014	Morrisons	140	Medium	Calib.	2022-04-30
UK015	Sainsburys	50	Small	Calib.	2022-04-26
UK016	Sainsburys	270	Large	Test	2022-04-21
UK017	Tesco	205	Large	Test	2021-06-14
UK018	Tesco	390	Large	Calib.	2022-04-21
UK019	Tesco	350	Large	Test	2022-04-30
UK020	Waitrose	220	Large	Calib.	2022-04-26
UK021	Waitrose	95	Medium	Test	2022-04-17
UK022	Waitrose	130	Medium	Test	2022-04-30
UK023	Tesco	410	Large	Test	2023-05-20
UK024	Tesco	250	Large	Calib.	2023-05-27
UK025	Lidl	120	Medium	Test	2022-04-30
UK026	Lidl	75	Medium	Test	2023-04-11
UK027	Aldi	120	Medium	Calib.	2022-06-16
UK028	Tesco	450	Large	Test	2021-04-20
UK029	Morrisons	490	Large	Calib.	2023-05-21
UK030	Sainsburys	400	Large	Test	2021-05-31

Appendix C: Per-site counting results

Per-site de-duplicated counts on the twenty held-out test sites for each model, the raw data behind the aggregate counting and occupancy metrics of Section 4.4. Truth is the hand-annotated, seam-de-duplicated count; each model’s count is produced at its own F_1 -optimal operating threshold.

Table 10: Per-site vehicle counts on the twenty held-out UK test sites. Cap. is the registry’s estimated capacity; Truth is the ground-truth count; the remaining columns are the predicted counts of the baseline, transfer and UK-adaptation models.

Site	Cap.	Truth	Baseline	Transfer	UK-adapt
UK002	180	141	286	277	144
UK003	95	17	8	8	15
UK004	130	8	8	2	8
UK005	260	105	19	151	109
UK006	105	17	10	41	19
UK007	250	104	183	158	106
UK008	130	71	34	154	73
UK011	400	93	190	205	97
UK012	140	24	4	25	24
UK013	100	93	76	159	79
UK016	270	100	343	263	114
UK017	205	63	100	161	68
UK019	350	51	52	91	41
UK021	95	22	10	34	22
UK022	130	103	4	224	103
UK023	410	146	313	313	136
UK025	120	64	75	24	67
UK026	75	14	7	13	10
UK028	450	138	336	330	162
UK030	400	139	86	263	144

Per-site entrance-proximity statistics for the twenty held-out test sites, the raw data behind the aggregates of Table 8. For each entrance type, r_{dist} is the distance ratio (Equation (8); < 1 means vehicles sit closer to the entrance than the average available parking space) and ρ_s the Spearman gradient (< 0 means relative fill declines with distance from the entrance).

Table 11: Per-site entrance-proximity statistics on the twenty held-out test sites, for the retail (pedestrian) and car-park (vehicle) entrances. Computed from the UK-adapted detector’s centroids at each site.

Site	Retail entrance		Car-park entrance	
	r_{dist}	ρ_s	r_{dist}	ρ_s
UK002	0.96	−0.41	1.05	+0.69
UK003	0.75	−0.44	0.89	−0.28
UK004	1.24	+0.59	0.62	−0.38
UK005	0.91	−0.10	1.00	+0.24
UK006	0.78	−0.30	0.92	−0.25
UK007	0.80	−0.54	1.01	+0.14
UK008	0.87	−0.46	1.08	+0.22
UK011	0.91	−0.03	0.91	−0.25
UK012	1.12	+0.70	0.89	−0.67
UK013	0.79	−0.92	1.17	+0.33
UK016	0.83	−0.46	1.15	+0.31
UK017	0.90	−0.25	1.07	+0.39
UK019	0.71	−0.53	0.76	−0.52
UK021	0.95	−0.12	1.32	+0.53
UK022	0.94	−0.11	0.97	−0.22
UK023	0.90	−0.30	1.04	+0.17
UK025	0.93	−0.33	0.92	+0.21
UK026	0.77	−0.77	0.76	−0.30
UK028	0.78	−0.63	0.94	−0.22
UK030	0.75	−0.60	0.98	+0.21
Mean	0.88	−0.30	0.97	+0.02

Appendix D: Software environment

The project used two environments (Table 4): an NVIDIA GPU server (Rocky Linux 9.8) for model training and inference, and a local macOS machine for data preparation, annotation and spatial analysis. Both were installed from the pinned `requirements.txt` below. The pins are identical across the two; on the Linux server `torch==2.3.0` resolved to the CUDA build (2.3.0+cu121, with CUDA 12.1 and cuDNN 8.9.2.26), whereas on macOS it resolved to the CPU/MPS build.

Listing 9: Pinned Python dependencies (`requirements.txt`).

```
1 # Deep learning + computer vision
2 torch==2.3.0
3 torchvision==0.18.0
4 ultralytics==8.2.0
5 opencv-python==4.9.0.80
6
7 # Geospatial - GDAL installed first via Homebrew:
8 # brew install gdal && pip install GDAL==3.13.0
9 rasterio==1.3.10
10 geopandas==0.14.4
11 shapely==2.0.4
12 fiona==1.9.6
13 pyproj==3.6.1
14
15 # Data analysis + visualisation
16 pandas==2.2.2
17 matplotlib==3.9.0
18 seaborn==0.13.2
19 numpy==1.26.4
20 pillow==10.3.0
21 folium==0.14.0
22
23 # Notebook
24 jupyterlab==4.2.0
25 ipykernel==6.29.4
```